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Experimental Measurement of Bargmann Invariants using a Nuclear Magnetic Resonance Quantum Computer

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Acknowledgments

This project stems from the initiative of the interest group Quantum@UC. This group is a multidisciplinary community made up of individuals interested in quantum technologies, with the shared goal of developing and exploring new possibilities in science and engineering in the fields of quantum computing, communication, and sensing at the University of Coimbra.

The Quantum@UC group emerged from the academic community's interest in quantum technologies and from the need to further develop this field at the University of Coimbra. In the Department of Physics and the Department of Electrical and Computer Engineering, optional courses in quantum technologies and quantum computing were created by Professor Helena Alberto and Professor Jorge Lobo, respectively. In collaboration with the Department of Informatics Engineering, the group was formed, with Professor Edmundo Monteiro appointed by the University's Rectorate to coordinate it. Since then, the group has organized several initiatives, including the creation of a non-degree course in quantum technologies for the 2024–2025 academic year [1].

The project itself is being developed at QuantumLab@UC, a new research laboratory at the University of Coimbra focused on quantum technologies, funded by the Recovery and Resilience Plan (PRR) and the European Union. The laboratory has two rooms, one in the Department of Electrical and Computer Engineering and the other in the Department of Physics, equipped with two NMR-based quantum computers, one with two qubits and the other with three qubits, as well as experimental Quantum Key Distribution (QKD) equipment for demonstrating the principles of quantum cryptography.

For the development of this project, I am being supervised by Professor Sagar Pratapsi and co-supervised by Professor Rui Vilão, both members of the Quantum@UC group.

The project arises as a continuation of the study carried out by Professor Sagar Pratapsi et al. on the characterization of Bargmann invariants [2], with the measurement of this quantity being the proposed procedure for testing the practical limitations of the NMR-based quantum computer.

Resumo

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Abstract

Quantum technologies offer new and promising alternatives across several areas of science. One of the most prominent fields is quantum computing, whose goal is to harness purely quantum phenomena to perform computational tasks faster and more efficiently than classical computers. In this project, I present, in a clear and detailed manner, the operation and use of an NMR-based quantum computer.

These computational tasks are expressed through quantum logic gate circuits, which must be implemented according to the physical system used as a platform. In this project, the implementation of a destructive $SWAP$-test gate is achieved by approximating the system's evolution when a radiofrequency magnetic field is applied, using computational methods to reproduce the desired operation. Its fidelity is then evaluated and compared with other similar implementations from the state of the art.

With well-characterized quantum gate circuits, it becomes possible to perform computational tasks such as prime factorization using Shor's algorithm or, in the case of this project, the measurement of Bargmann invariants using $SWAP$-test and *cycle*-test circuits. These are fundamental quantities for characterizing the corresponding quantum states, allowing one to determine their equivalence or detect the presence of quantum entanglement.

"Intelligence is the ability to avoid doing work, yet getting the work done"

Linus Torvald

Contents

List of Acronyms

NMR Nuclear Magnetic Resonance

GRAPE Gradient Ascent Pulse Engineering

L-BFGS Limited-memory BFGS

BFGS Broyden–Fletcher–Goldfarb–Shanno algorithm

ADAM Adaptive Moment Estimation

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1 Introduction

1.1 Motivation

Over the past three decades, quantum technologies have experienced explosive growth, driven by fundamental advances in quantum mechanics and experimental quantum control. Today, it is possible to manipulate, control, and measure individual quantum systems with remarkable precision, enabling a wide range of emerging technologies.

Among these are quantum computing, quantum communication and cryptography, quantum simulation, quantum metrology, and quantum sensing. Although these fields target different applications, they all make use of uniquely quantum phenomena, such as superposition and entanglement, to approach certain tasks in ways that are classically impossible. The present work is situated within the field of quantum computing, whose objective is to harness these quantum effects to implement computational algorithms that are more efficient than their classical counterparts.

The experimental platform considered in this work is a liquid-state Nuclear Magnetic Resonance (NMR) quantum computer that provides direct low-level access to the radio-frequency control pulses that manipulate the qubits. This allows for custom quantum gates to be designed and experimentally implemented, making the platform particularly valuable both as a research tool and as an educational resource, enabling the connection between the theoretical description of quantum algorithms and their physical realization.

From an educational perspective, an in-depth understanding of the operation of the NMR quantum computer provides students with practical experience in quantum control, pulse sequence design, and experimental quantum information processing. From a research perspective, the platform enables the development and validation of new control techniques, quantum algorithms, and experimental demonstrations in areas including quantum computing, experimental physics, theoretical physics, and molecular chemistry.

For these reasons, the characterization of the quantum computer's operation must naturally be the first step. Understanding the performance and limitations of the system is essential for determining which quantum operations can be implemented reliably and for identifying the sources of experimental error that limit gate performance. Such characterization also provides a quantitative basis for evaluating future improvements in pulse optimization and hardware calibration.

In this work, this characterization is carried out using *Bargmann invariants* as a benchmarking tool. Since Bargmann invariants depend on the relative geometric phase accumulated by multiple quantum states, they are particularly sensitive to imperfections in quantum operations while remaining independent of global phase. Their measurement therefore provides a strict experimental test of the accuracy of implemented control pulses and offers a practical metric for assessing the performance of the NMR quantum computing platform.

1.2 State of the Art

1.2.1 Quantum computation in NMR

Nuclear Magnetic Resonance (NMR) was one of the earliest experimental platforms for quantum computing. Research in this area began around 1997, and it was shown to satisfy the conditions later formalized as the DiVincenzo criteria for quantum computation [3, 4]. The criteria, and the reasons why they are satisfied, are as follows:

1. A scalable system with well-characterized qubits
   NMR systems use spin-1/2 nuclei to represent each qubit. The molecules containing the qubits are placed in a liquid solution, to which a strong constant magnetic field is applied. The splitting of the energy levels due to the Zeeman effect provides the logical states $|0\rangle$ and $|1\rangle$.

2. Ability to initialize the qubits to a known fiducial state
   In other technologies, initialization is commonly achieved by cooling the system to its ground state, $|000\ldots\rangle$. In liquid-state NMR, such cooling is not possible because the sample must remain *in the liquid state*. The alternative is to initialize the system in a known pseudo-pure state, which can be treated as a pure state of the form $|000\ldots\rangle$, but with only a significantly smaller fraction of the total population of the sample.

3. Long decoherence times

   According to DiVincenzo, the fastest relaxation time should be 10^4 to 10^5 times longer than the operation times of the phenomena used as quantum gates [3]. In liquid-state NMR, decoherence times are on the order of seconds, while the operation times of quantum gates, implemented through interactions between different intramolecular spins, are on the order of microseconds, thus satisfying this criterion.

4. Implementation of a universal set of quantum gates

   In NMR implementations, through single-spin rotations about axes in the xy plane of the Bloch sphere, together with periods of free evolution under coupling interactions, it is possible to approximate any desired evolution or operation on the spins of the system. This makes the definition of a universal set of gates possible.

5. Ability to measure each specific qubit

   The measurement of the spin state is obtained from the NMR spectrum, resulting from the application of the corresponding excitation pulses. Considering that the sample is generally at room temperature, what is obtained is an average over the spin states of each molecule in the ensemble, represented by a density matrix.

For several years, NMR-based quantum computing led experimental development in the field, culminating in the first implementation of Shor's algorithm, carried out on a 7-qubit NMR system [5]. This marked an important milestone on the path toward demonstrating quantum supremacy.

However, progress slowed as the intrinsic limitations of the technique, already anticipated at the beginning of its development [6], began to be reached. The main limiting factor is its severely restricted scalability. To use a system with a larger number of qubits, one requires a molecule in a liquid-state solution containing the same number of spin-active nuclei, with sufficiently large energy-level separations to allow individual distinction and addressing. As a result, viable systems are limited to approximately 12 to 15 qubits, which is insufficient for more complex circuits and operations.

In addition, these systems require initialization into a pseudo-pure state, since the sample cannot be cooled to reach a true pure ground state. Because only a small fraction of the total sample population is effectively used, the intensity of the output signal is further reduced, once again making individual distinction and addressing in more complex systems more difficult.

At present, NMR quantum computing has been surpassed in speed and precision by trapped-ion systems [7, 8], and in terms of scalability by superconducting platforms [9, 10]. Today, the value of the platform lies in its accessibility for research and the testing of theoretical models, since pulse modulation for controlling quantum states is already a well-studied subject and shares several aspects analogous to quantum control techniques used in other quantum computing platforms.

1.2.2 Implementation of the Fredkin quantum gate/ swap test

swap test[11]

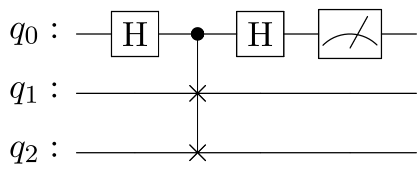

Figure 1.1: Conventional SWAP test

The controlled-SWAP gate, or Fredkin gate, was first defined by Edward Fredkin and Tommaso Toffoli in their paper *"Conservative Logic"* [12]. The authors proposed a new methodology for classical computation, grounded in microscopic physical principles, aiming to improve the efficiency and performance of computational processes by introducing concepts such as reversible logic and conservative logic. They presented a paradigm shift from computation based on macroscopic physical axioms to a model in which computational processes more directly reflect the mechanisms of the underlying physical systems, such as the conservation of information and energy. Since quantum computing shares these same fundamental principles, the quantum implementation of the Fredkin gate, as well as the Toffoli gate, can be seen as a natural step in the development of quantum logic gates.

The Fredkin gate is a three-qubit gate that swaps the states of the two target qubits, conditioned on the state of the control qubit. It is a reversible gate, meaning that the information contained in the input signals can be fully recovered from the output signals, in accordance with the principles of conservative logic: information preservation and minimization of dissipated energy. Furthermore, when combined with the Hadamard quantum gate, it forms a universal set of quantum gates, allowing any computational operation to be decomposed into a circuit composed of just these two logical gates.

Several proposals and implementations of the Fredkin gate have been made across multiple quantum computing platforms. The first approach involved decomposing the gate operation into a set of two-qubit gates. This was demonstrated in 1996 and later proven, in 2015, to be optimized, with the minimum number of gates required being five [13, 14]. From this perspective, theoretical proposals were made for implementing the Fredkin gate in photonic quantum systems [15, 16], with the first experimental demonstration occurring in 2016. This demonstration introduced a control qubit into the unitary operation that swaps the states of the pair of target qubits, resulting in the controlled-SWAP operation [17].

A second approach, aimed at minimizing unnecessary operations, computational errors, and energy dissipation/costs, especially with large-scale quantum computing in mind, focused on implementing the Fredkin gate in a single step. In this approach, a single control Hamiltonian is applied to the three-qubit system to directly realize the desired logical operation. Theoretical proposals from 2012 and 2020 suggest using the quantum Zeno effect in systems of atoms trapped in optical cavities, to prevent unwanted transitions through frequent measurements and thereby simplify the resulting Hamiltonian [18, 19].

Finally, in NMR-based systems, this second approach is implemented by optimizing the parameters of control pulse sequences, which effectively corresponds to applying a control Hamiltonian to the spin-bearing atoms to approximate the system's evolution to the desired Fredkin gate operation [20].

1.2.3 Bargmann Invariants

Bargmann invariants were introduced by Valentine Bargmann in his analysis of Wigner's theorem on symmetry operations in quantum systems [21]. They are gauge-invariant quantities defined through cyclic overlaps of quantum state vectors. Represented as complex numbers, they are independent of any arbitrary choice of reference phase, allowing for the direct assessment of properties of the quantum state space itself [2, 22].

Their measurement is therefore useful both for the fundamental characterization of a given quantum state and for practical applications in quantum physics and quantum information science. Examples include comparing two quantum states to verify their local equivalence or detecting quantum entanglement between states without relying on full quantum state tomography [23].

Measurement of Bargmann invariants, for both pure and mixed quantum states, can be performed using *cycle*-tests, which are based on $SWAP$-tests, employing Hadamard gates and multiple Fredkin gates to swap and measure the quantum states [24].

1.2.4 SPINQ Gemini Lab

Para acrescentar contexto à informação que se segue, o sistema físico consiste num par de ímanes estáticos de neodímio que criam um campo magnético intenso de 0.65 T, mantido constante e homogéneo por shimming. A amostra encontra-se num pequeno tubo de ensaio inserido no centro do campo magnético, e envolvido por uma bobina utilizada para a criação dos pulsos de radiofrequência de controlo por indução eletromagnética.

1.3 Objective

Experimental measurement of Bargmann invariants, as described by Sagar Pratapsi et al. [2], using a three-qubit NMR-based quantum computer, through the implementation of $SWAP$-tests and *cycle*-test circuits, employing Hadamard and Fredkin gates for this purpose.

This project aims to evaluate the practical limitations of experimentally measuring Bargmann invariants in quantum systems based on NMR technology. In addition, it seeks to develop a performance characterization tool that is both useful and practical for quantum computers, extending beyond the specific three-qubit machine used in this work.

2 Liquid State NMR

2.1 Baseline concepts

2.1.1 Qubit

A quantum bit, or *qubit*, is the fundamental unit of quantum information. Physically, it is a quantum-mechanical two-level system, which can be realized in a variety of ways. For example, as the spin of an electron or nucleus that can occupy either a spin-up or spin-down state, or the polarization of a photon that can be horizontal or vertical.

Unlike its classical counterpart, the bit, which can only assume one of the two states 0 or 1, a qubit may exist in a coherent superposition of both basis states. Because of this, the outcome of a measurement is inherently probabilistic, with probabilities determined by the state's probability amplitudes. This ability to exploit superposition, together with other uniquely quantum phenomena such as interference and entanglement, enables quantum algorithms that can outperform their classical counterparts for specific computational problems.

The state of a qubit can be represented in several equivalent ways. One is as a linear combination of the orthonormal computational basis states $|0\rangle$ and $|1\rangle$, where the complex amplitudes satisfy the normalization condition $|\alpha|^2 + |\beta|^2 = 1$

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle. \tag{2.1}$$

Alternatively, any single-qubit state can be represented geometrically on the **Bloch sphere**, by a state vector $\vec{r}$. For a pure state, the state vector has a norm of one and can be parameterized by two angles, θ and ϕ, as shown in Figure 2.1. The state vector can be expressed in terms of these angles as

$$|\psi\rangle = \cos\frac{\theta}{2}|0\rangle + e^{i\phi}\sin\frac{\theta}{2}|1\rangle. \tag{2.2}$$

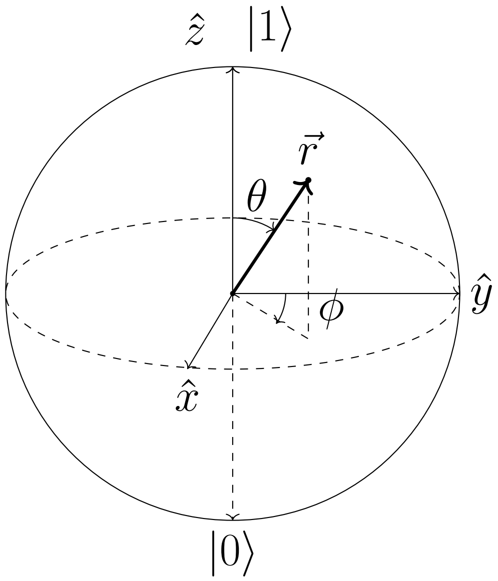

Figure 2.1: Representation of a single-qubit state on the Bloch sphere.

2.1.2 Unitary operation

A unitary operation is a transformation that is perfectly reversible. For every unitary operation described by a matrix U, there exists an inverse matrix $U^\dagger$ that describes exactly the undoing of the operation. Consequently, multiplying these two matrices must always equal the identity matrix.

$$UU^\dagger = U^\dagger U = I \tag{2.3}$$

A unitary operation also preserves the norm of the quantum state vector, and therefore the sum of collapse probabilities. This guarantees that no information is lost throughout the process.

$$\langle x, y\rangle = \langle Ux, Uy\rangle \tag{2.4}$$

Because of these characteristics, quantum logic gates should aim to implement as close to an unitary operation as possible.

2.1.3 Density matrix

2.2 The choice of molecule

In liquid-state NMR quantum computing, the computational platform consists of an ensemble of molecules containing spin-$\frac{1}{2}$ nuclei coupled through direct or indirect spin-spin interactions. The molecular structure determines both the number of available qubits and the interactions between them, thereby defining the strategies used for control and quantum information processing.

A key distinction is whether the molecule forms a heteronuclear or homonuclear spin system. In a heteronuclear system, each qubit corresponds to a nucleus of a different chemical species, whereas in a homonuclear system two or more qubits are nuclei of the same species. Since the number of suitable spin-$\frac{1}{2}$ nuclei is limited, purely heteronuclear systems are generally restricted to a small number of qubits, while larger systems typically contain one or more homonuclear subsystems.

Table 2.1: Spin-$\frac{1}{2}$ nuclei commonly used in NMR quantum computing.

Nucleus	**Natural abundance (%)**	**Gyromagnetic Ratio ($\gamma/2\pi$, MHz/T)**
^{1}H	99.99	42.58
^{13}C	1.07	10.71
^{15}N	0.37	−4.32
^{19}F	100	40.05
^{29}Si	4.68	−8.47
^{31}P	100	17.24
^{77}Se	7.63	8.13

Heteronuclear systems greatly simplify individual qubit addressing because each nucleus possesses a distinct Larmor frequency, allowing selective excitation with radio-frequency pulses. For this reason, they are generally preferred whenever the desired number of qubits permits their use.

2.2.1 Dimethyl phosphite

The quantum computing platform employed in this work is the heteronuclear molecule dimethyl phosphite, $(CH_3O)_2POH$, dissolved in a liquid solution within the NMR quantum computer. The two qubits are encoded in the directly bonded ^{31}P and ^{1}H nuclei located at the center of the molecule, as illustrated in Figure 2.2.

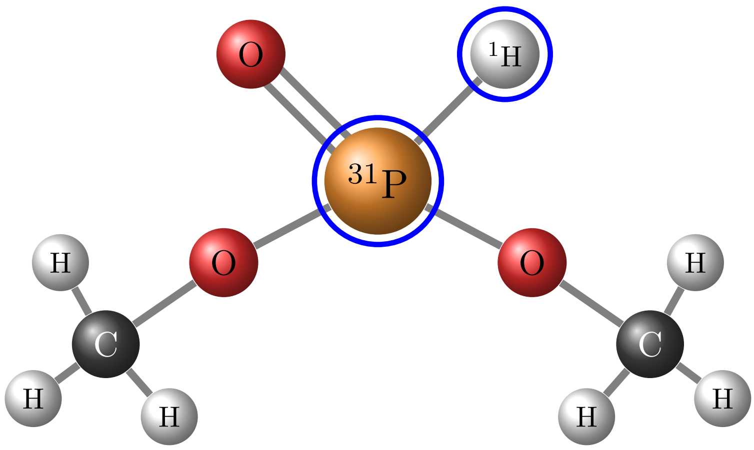

Figure 2.2: Dimethyl phosphite molecular structure. Nuclei used as qubits are highlighted in blue.

3 Quantum Computing with NMR

In this chapter, I present the underlying physical principles and describe the processes that justify the use of NMR for quantum computing. The discussion follows the logical sequence of the DiVincenzo criteria [3] for the construction of a quantum computing system, as they provide a simple and coherent framework for introducing quantum computation based on NMR.

3.1 DiVincenzo 1: Qubit caracterization

3.1.1 Isolated qubit

Each qubit is represented by a spin $S = 1/2$ nucleus which, when subjected to a uniform magnetic field, exhibits two energy states depending on whether it is aligned or anti-aligned with the direction of the magnetic field. In isolation, taking $\hat{z}$ as the direction of the constant magnetic field $\vec{B_0}$, the time evolution of the nucleus is described by

$$H_z = -\vec{\mu} \cdot \vec{B_0} = -\frac{\gamma\hbar}{2} B_0 \vec{\sigma} \cdot \hat{z} = -\hbar\omega_0 I_z. \tag{3.1}$$

The time evolution operator $U = e^{-i\frac{H_z}{\hbar}t}$ describes a precession of the state vector $\vec{r}$ on the Bloch sphere about the axis of the magnetic field $\vec{B_0}$ at the Larmor frequency, ω_0. For the experimental magnetic field of 0.65 T, the frequencies are on the order of tens of MHz.

When aligned with $\hat{z}$, $m_s = 1/2$, the nuclear spin is in the stable lower-energy state: $|\uparrow\rangle$ or $|0\rangle$; when anti-aligned, $m_s = -1/2$, it is in the unstable higher-energy state: $|\downarrow\rangle$ or $|1\rangle$. This lifting of the energy-level degeneracy is known as the Zeeman effect, with an energy splitting between the levels equal to $\hbar\omega_0$.

In heteronuclear systems, the spectral distinction between nuclei is straightforward because each nucleus has a different gyromagnetic ratio, γ, and therefore a different Larmor frequency. In homonuclear systems, distinction is achieved through the chemical shifts, δ_i, of the nuclei within the molecule, which modify the Larmor frequency of each nucleus. These shifts arise from the shielding effect of the electronic cloud against the external magnetic field, which depends on the molecular geometry, and are typically on the order of tens of kHz.

3.1.2 Coupling effects

Within a molecule, there are two types of intramolecular internuclear interactions. The first is the direct magnetic interaction between nuclear dipoles, described by

$$H_D = \sum_{i<j} \frac{\mu_0 \gamma_i \gamma_j \hbar}{4\pi |\vec{r_{ij}}|^3} \left[\vec{I^i}\vec{I^j} - \frac{3}{|\vec{r_{ij}}|^2} (\vec{I^i} \cdot \vec{r_{ij}})(\vec{I^j} \cdot \vec{r_{ij}}) \right]. \tag{3.2}$$

After several approximations and simplifications, its contribution to the qubit state dynamics in an ensemble of molecules in liquid solution approaches zero due to rapid molecular tumbling and symmetry considerations.

The second is the spin–spin interaction, J-coupling, arising from interactions between the nuclei and the local electrons forming their molecular bonds. This interaction is described by

$$H_J = \sum_{i<j}^{n} 2\pi\hbar J_{ij} (I_x^i I_x^j + I_y^i I_y^j + I_z^i I_z^j). \tag{3.3}$$

The J factor is a frequency difference between energy levels of the fine structure created by this interaction, and it depends on the nuclear species involved, typically being on the order of up to hundreds of Hz. The expression above can be simplified when $|\omega_i - \omega_j| \gg 2\pi J_{ij}$ to

$$H_J = \sum_{i<j}^{n} 2\pi\hbar J_{ij} I_z^i I_z^j. \tag{3.4}$$

Adding the static field term, the Hamiltonian for any system of n coupled nuclear spins therefore takes the approximate form of

$$H_0 = \sum_{i}^{n} 2\pi\hbar\omega_0^i I_z^i + \sum_{i<j}^{n} 2\pi\hbar J_{ij} I_z^i I_z^j. \tag{3.5}$$

3.1.3 Control Hamiltonian

The state of the qubit can be controlled by applying an external rotating magnetic field $\vec{B_1}(t)$, with components along $\hat{x}$ and $\hat{y}$, whose rotation frequency, ω_{rf}, is resonant with the qubit's Larmor frequency, ω_0.

In practice, the magnetic control field applied by the coil is linearly polarized. However, this field can be decomposed into two counter-rotating magnetic fields. One of them rotates in the same direction as the qubit precession and can be brought into resonance by tuning the frequency ω_{rf}. The other, rotating in the opposite direction, is far off-resonance and can, to first approximation, be neglected. Nevertheless, its presence introduces a small correction to the Larmor frequency, known as the *Bloch–Siegert shift*.

Let $\hat{z}$ be the direction of the constant magnetic field, as previously described, and $\hat{x}$ the direction of the axis of the unidirectional field generated by the control coil. The evolution of the qubit state can then be described by the following time-dependent Hamiltonian in equation 3.8,

$$H_z = -\vec{\mu} \cdot \vec{B_0} = -\frac{\gamma\hbar}{2} B_0 \vec{\sigma} \cdot \hat{z} = -\hbar\omega_0 I_z \tag{3.6}$$

$$H_x = -\vec{\mu} \cdot \vec{B_1} = -\frac{\gamma\hbar}{2} B_1 \cos{(\omega_{rf} t + \phi_{rf})} \vec{\sigma} \cdot \hat{x} = -\hbar\omega_1 \cos{(\omega_{rf} t + \phi_{rf})} I_x \tag{3.7}$$

$$H_{qubit} = -\hbar\omega_0 I_z - \hbar\omega_1 \cos(\omega_{rf} t + \phi_{rf}) I_x \tag{3.8}$$

where $\vec{\mu} = \gamma\hbar\vec{\sigma}/2$ is the nuclear magnetic moment, γ is its gyromagnetic ratio, and $\vec{\sigma}$ is the vector of Pauli matrices. B_0 and B_1 are the magnitudes of the constant magnetic field and the field generated by the control coil, respectively. The Larmor frequency $\omega_0 = \gamma B_0$ and the Rabi frequency $\omega_1 = \gamma B_1$ are distinct and crucial for describing and controlling the state of the system. The term corresponding to the J-coupling is omitted because it arises from a constant interaction that cannot be switched off, but only counteracted.

The time evolution of the qubit state is obtained by solving the Schrödinger equation for this Hamiltonian.

$$i\hbar \frac{\partial |\psi(t)\rangle}{\partial t} = H_{qubit} |\psi(t)\rangle \tag{3.9}$$

A transformation to a reference frame rotating with the precession about $\hat{z}$ at frequency ω_{rf} simplifies the final result by removing the explicit time dependence. Terms in the Hamiltonian with rapidly oscillating frequencies are also neglected under the *Rotating Wave Approximation*, as they average out over multiple oscillations, leading to a simplified control Hamiltonian in equation 3.11.

$$|\psi^{rot}(t)\rangle = e^{-i\omega_{rf} t I_z}|\psi(t)\rangle \tag{3.10}$$

$$H^{rot} = -\hbar(\omega_0 - \omega_{rf})I_z - \frac{\hbar\omega_1}{2}[\cos(\phi_{rf})I_x - \sin(\phi_{rf})I_y] \tag{3.11}$$

The full system Hamiltonian is then

$$H_0 = \underbrace{\sum_{i=1}^{n} 2\pi\hbar(\omega_0^i - \omega_{rf})I_z^i}_{\text{Constant magnetic field}} + \underbrace{\sum_{i<j}^{n} 2\pi\hbar J_{ij} I_z^i I_z^j}_{J\text{-coupling}} + \underbrace{\sum_{i=1}^{n} 2\pi\hbar\omega_1^i \left[\cos(\phi_{rf})I_x^i + \sin(\phi_{rf})I_y^i\right]}_{\text{Control RF field}}. \tag{3.12}$$

In this Hamiltonian, the parameters that can be controlled to manipulate the qubit state are:

1. The rotation frequency of the reference frame, ω_{rf}, typically kept in resonance with the Larmor frequency ω_0, eliminating the first term of the Hamiltonian, but which can be adjusted via detuning to modify the outcome of the applied signal;
2. The phase, ϕ_{rf}, which controls the rotation axis direction between $\hat{x}$ and $\hat{y}$;
3. The Rabi frequency, ω_1, directly proportional to the intensity of the calibrated control magnetic field, which determines the speed of rotation.

The calculation of the time evolution under this Hamiltonian reveals the fourth and final control parameter: the duration of application of the Hamiltonian, t.

$$|\psi(t)\rangle = \exp\Big[-i\hbar(\omega_0^i - \omega_{rf})tI_z^i\Big]\exp\Big[-i\hbar J_{ij}tI_z^i I_z^j\Big]\exp\Big[-i\hbar\frac{\omega_1}{2}t\,(\cos\phi_{rf}\,I_x - \sin\phi_{rf}\,I_y)\Big]|\psi(0)\rangle \tag{3.13}$$

3.2 DiVincenzo 2: Initialization to a pseudo-pure state

Once the qubit system can be controlled, the next requirement for performing computation is the preparation of the initial state in which the operations will be carried out, since only by knowing the initial state does the result of the applied algorithm become meaningful.

The most common approach across quantum computing platforms consists of cooling the system to temperatures close to absolute zero, so that all qubits occupy the lowest-energy state, $|0\rangle$, thereby establishing a well-defined initial state. The techniques used vary depending on the underlying technology. In the case of superconducting qubits, dilution refrigerators based on helium are used. In trapped-ion systems, cooling is performed using lasers: electromagnetic fields are employed to reduce the ions' kinetic energy until they reach the motional ground state.

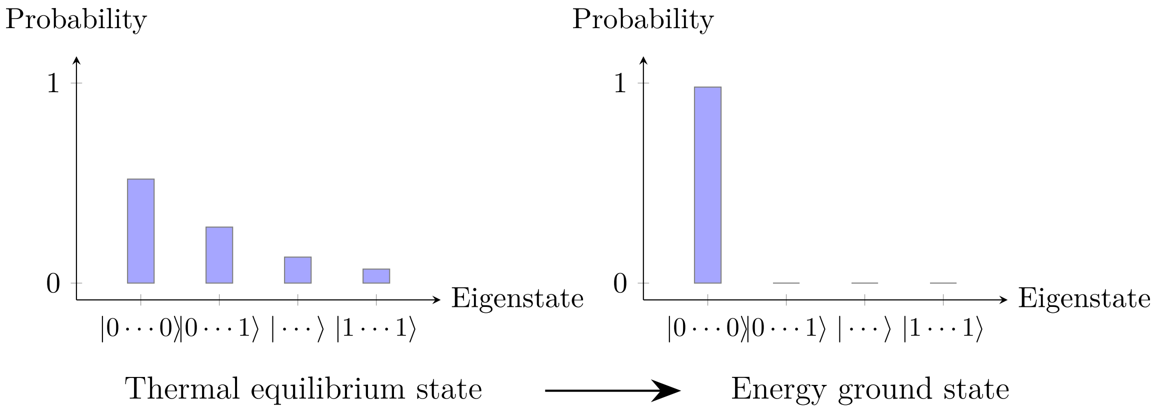

Figure 3.1: Illustration of quantum system initialization from a thermal equilibrium state to the ground state.

3.2.1 Pseudo-pure state

In the present case of liquid-state NMR, it is essential that the sample remains in the liquid phase, since both the control and measurement techniques rely on phenomena and effects that occur only in the room-temperature regime. In the liquid state, the molecules in the sample undergo rapid and random motion, so that small magnetic fluctuations are effectively averaged out.

For this reason, the system can be treated as an ensemble of resonant nuclear spins, enabling both frequency-selective control using radio-frequency pulses and state measurement through its narrow, well-defined spectral lines.

In NMR, initialization is achieved through a pseudo-pure state, a system state of the qubits that behaves like a pure state under any applied operation. The goal is to equalize the populations of the $2^n - 1$ excited states, leaving the population of the ground state, that dominates in the system's thermal equilibrium, unchanged.

The resulting pseudo-pure state can be decomposed into two components,

$$\rho = (1 - \epsilon)\frac{\mathbb{I}}{4} + \epsilon|00\rangle\langle 00|. \tag{3.14}$$

1. Maximum mixed state: all possible states of the system have equal population. This component does not change under unitary operations, since it is equivalent to the identity matrix, nor is it detected in the NMR spectrum.

2. Effective pure state: population is present only in the ground state of the system; it behaves exactly like the desired pure ground state, but it represents only a fraction of the total sample population, thereby reducing the intensity of the NMR spectral signal.

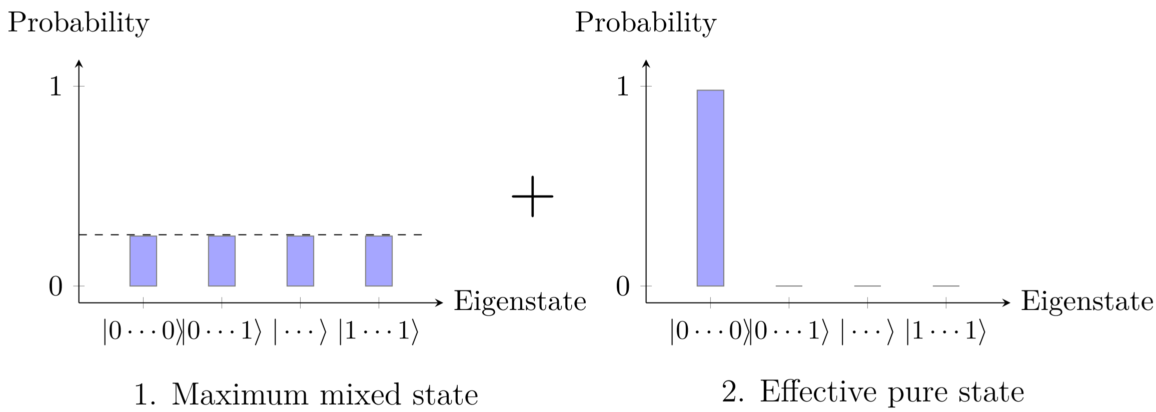

Figure 3.2: Probability distributions of each component of a pseudo-pure state. The left plot shows the maximum mixed state, in which all states have equal population. The right plot shows the effective pure state, in which only the ground state has population.

This limitation of quantum computing in liquid-state NMR of having to resort to pseudo-pure states is the reason why it is not possible to scale the number of qubits in this platform beyond around ten qubits. The signal intensity decreases exponentially with the number of qubits, making it impossible to measure the result of the computation. The pure fraction of the pseudo-pure state is the state polarization, ϵ, given by

$$\epsilon = \frac{2\sinh\left(nh\omega/2k_BT\right)}{2^n\cosh^n\left(h\omega/2k_BT\right)}. \tag{3.15}$$

In the high-temperature limit, such as at room temperature, where $h\omega \ll k_BT$, each additional qubit approximately halves the NMR signal.

$$\epsilon \approx \frac{nh\omega/2k_BT}{2^n} \approx n2^{-n} \tag{3.16}$$

As an example, a 100-qubit system at room temperature would produce a signal approximately 28 orders of magnitude below the background noise arising from thermal magnetization, making it impossible to measure. As mentioned previously, cooling the sample to reach a regime in which the signal is not significantly affected, which would be typically around 1 K, depending on the sample, is not feasible, since dipolar couplings re-emerge and spectral lines broaden, among other effects.

3.2.2 Preparation methods

There are three main methods for preparing a pseudo-pure state. The first, **Logical labeling with an auxiliary qubit**, makes use of a subset of energy states within the total system whose populations correspond to those required for the pseudo-pure state. In a 3-qubit system, for example, using an auxiliary qubit, the logical states $|00\rangle$, $|01\rangle$, $|10\rangle$, and $|11\rangle$ of a 2-qubit system are mapped onto four of the eight possible physical states of the 3-qubit system, with these four states sharing the same spin state in the auxiliary qubit, which acts as a label.

The result is a 2-logical-qubit system that exhibits a population distribution corresponding to a pseudo-pure state, to which any logical operations can be applied, provided that the spin of the auxiliary label qubit is not affected. This method is used only in homonuclear systems, since the natural distribution of state populations must follow specific patterns similar to those of a pseudo-pure state in order for them to be labelable.

The second, **Temporal averaging by permutation**, requires the same algorithm to be run for $2^n - 1$ initial states, in which the distribution of the excited states is cyclically permuted. The permutation of the excited states is performed by a pulse that repeatedly implements the unitary operation P_n:

$$P_n = \begin{bmatrix} 1 & 0 & 0 & 0 & \cdots & 0 & 0 \\ 0 & 0 & 0 & 0 & \cdots & 0 & 1 \\ 0 & 1 & 0 & 0 & \cdots & 0 & 0 \\ 0 & 0 & 1 & 0 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & 0 & \cdots & 1 & 0 \end{bmatrix} \tag{3.17}$$

The results are then averaged over the multiple runs. Since the unitary operations that compose the algorithms and the final measurement operations are linear processes, this average is equivalent to the result of the algorithm applied to the average of the initially prepared states.

Finally, the third method, **Spatial averaging**, is based on adjusting the relative populations of the states by exciting the system, rearranging the populations so that they resemble those of a pseudo-pure state, followed by the use of magnetic field gradients to remove unwanted off-diagonal elements of the density matrix.

The following initialization sequence, adapted from Jones [25], uses a series of rotations to redistribute the state populations, starting from a thermal equilibrium state. This is followed by either a magnetic field gradient that removes the off-diagonal elements of the density matrix, aptly named *crush* gradient, or a period of free evolution under the J-coupling interaction of duration $1/(2J)$. The resulting state is a pseudo-pure state.

$$\begin{aligned} I_z + S_z &\xrightarrow{60^\circ S_x} I_z + \frac{1}{2}S_z - \frac{\sqrt{3}}{2}S_y \\ &\xrightarrow{\text{crush}} I_z + \frac{1}{2}S_z \\ &\xrightarrow{45^\circ I_x} \frac{1}{\sqrt{2}}I_z - \frac{1}{\sqrt{2}}I_y + \frac{1}{2}S_z \\ &\xrightarrow{\text{J}-\text{coupling}} \frac{1}{\sqrt{2}}I_z + \frac{1}{\sqrt{2}}2I_xS_z + \frac{1}{2}S_z \\ &\xrightarrow{45^\circ I_{-y}} \frac{1}{2}I_z - \frac{1}{2}I_x + \frac{1}{2}2I_xS_z + \frac{1}{2}S_z + \frac{1}{2}2I_zS_z \\ &\xrightarrow{\text{crush}} \frac{1}{2}I_z + \frac{1}{2}S_z + \frac{1}{2}2I_zS_z \end{aligned} \tag{3.18}$$

Spatial averaging is the most popular method because it does not require additional auxiliary qubits, does not increase the complexity of logical gate design, and is faster than the temporal averaging procedure.

3.2.3 Method employed by Gemini

The method used for preparing the initial pseudo-pure state in the quantum computer employed in this work is an adaptation of the temporal averaging method. It uses a permutation operation on the excited states followed by a period of longitudinal relaxation of the system, during which the spins are allowed to evolve freely. Repeating this process cyclically causes the populations of the excited states to stabilize at similar values, resulting in a high-fidelity pseudo-pure state.

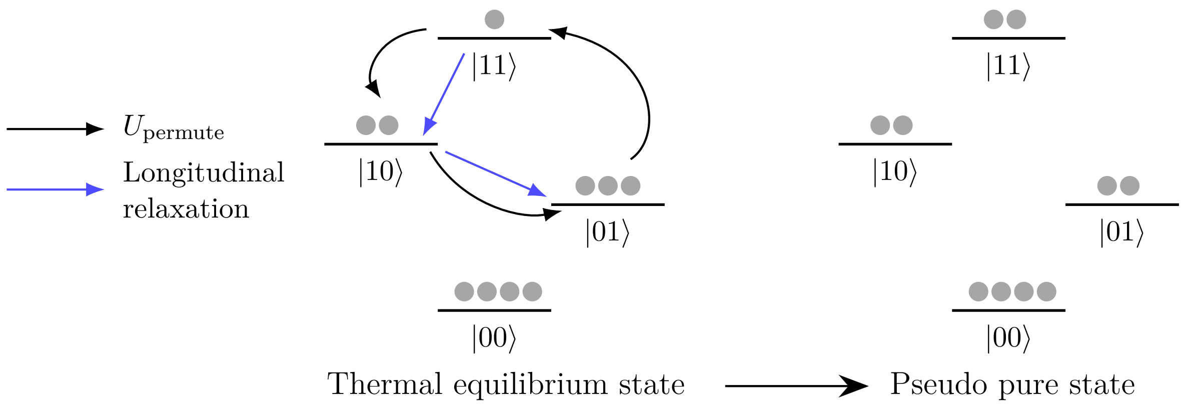

Figure 3.3: Initialization process permformed by Gemini to achieve a pseudo-pure state. The populations of the three excited states are cyclically permuted, after which a period of free evolution under solely the J-coupling interaction is performed. Over time, the populations of the excited states are equalized. Image adapted from SPINQ Gemini Lab manual.

3.3 DiVincenzo 3: Long Decoherence times

The main aspect that distinguishes a classical computer from a quantum one is the ability of qubits to exist in a state of superposition, enabling the processing of more information more rapidly and being essential for more complex algorithms. However, superposition cannot be maintained indefinitely. Both experimental measurements and interactions with the external environment cause a quantum state to lose coherence over time.

The coherence time of a state depends on the architecture used for the quantum computer and corresponds to the maximum time allowed for the implementation of algorithms. Naturally, quantum logic gates must have durations limited by these values so that they can act before the onset of decoherence.

3.3.1 Longitudinal Relaxation

Through spin-lattice relaxation, the longitudinal component of the nuclear spin, parallel to the direction of the magnetic field, relaxes from a higher-energy state toward the thermodynamic equilibrium state via interaction with the environment. The expected value of $\langle S_z \rangle$ evolves according to the exponential law

$$\langle S_z \rangle(t) = \langle S_z \rangle_0 + (\langle S_z \rangle_{eq} - \langle S_z \rangle_0)(1 - e^{-\frac{t}{T_1}}). \tag{3.19}$$

In this expression, $\langle S_z \rangle_0$ is the initial state of the longitudinal spin component before relaxation,

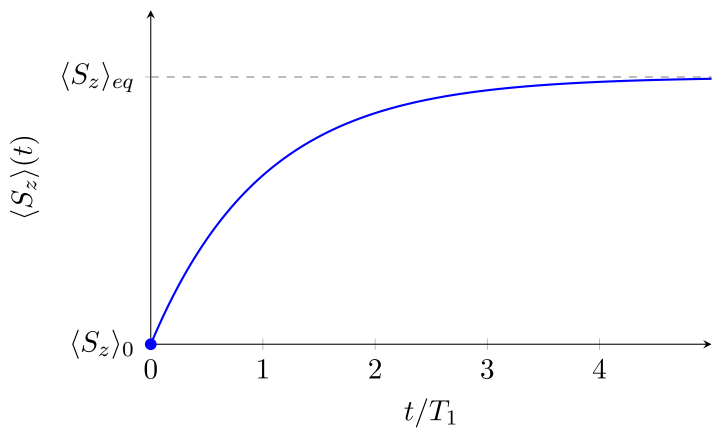

Figure 3.4: Evolution of $\langle S_z \rangle(t)$. The longitudinal component of the nuclear spin behaves as a magnetic dipole that relaxes toward an equilibrium state, which in this case is the alignment with the external magnetic field.

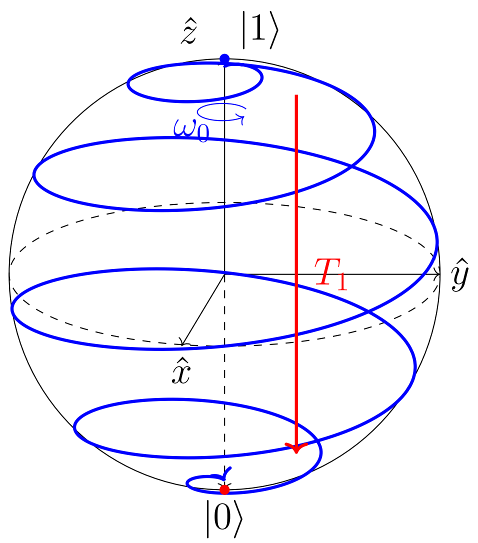

Figure 3.5: Bloch sphere representation of Longitudinal relaxation. While the longitudinal component of the nuclear spin relaxes from the excited state $|1\rangle$ toward the equilibrium state $|0\rangle$, the transverse component precesses around the external magnetic field at the Larmor frequency.

and $\langle S_z \rangle_{eq}$ is the component corresponding to maximum decoherence, i.e., the equilibrium state. The characteristic constant T_1 determines the rate of the relaxation process, with larger values of T_1 corresponding to slower relaxation and therefore more time available to implement the desired algorithms.

Its measurement can be easily performed by repeatedly exciting each qubit from the $|0\rangle$ state to the $|1\rangle$ state, allowing it to evolve freely, and measuring the $\langle S_z \rangle$ component of the state at different time intervals. The curve 3.19 is then fitted to the set of results, yielding a value for T_1.

3.3.2 Transverse Relaxation

Through spin–spin relaxation, the transverse component of the nuclear spin relaxes toward the equilibrium state, leading to the loss of phase information and of the state's superposition. This component, $\langle S_{xy} \rangle$, similarly evolves according to the exponential law

$$\langle S_{xy} \rangle(t) = \langle S_{xy} \rangle_0 e^{-\frac{t}{T_2}}. \tag{3.20}$$

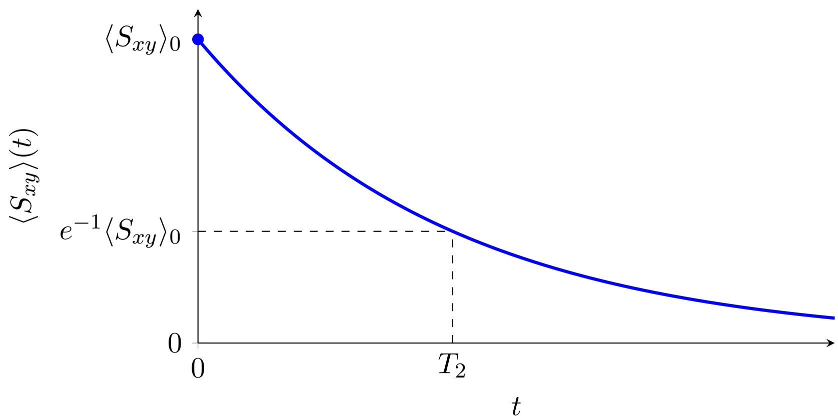

Figure 3.6: Evolution of $\langle S_{xy} \rangle(t)$. The transverse component of the nuclear spin relaxes toward the equilibrium state, leading to the loss of information. The relaxation rate is characterized by a time constant: T_2.

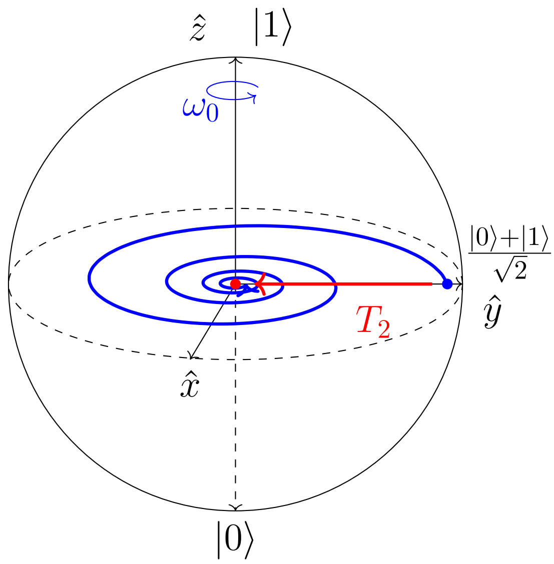

Figure 3.7: Bloch sphere representation of Transverse relaxation. The transverse component of the nuclear spin relaxes from an example superposition state $\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$, toward the equilibrium maximum mixed state, losing phase information and coherence. The precession around the external magnetic field continues at the Larmor frequency.

The relaxation rate is, similarly to longitudinal relaxation, characterized by a time constant T_2. T_2 relaxation is generally faster than T_1 relaxation, and is therefore the limiting factor.

As with T_1, the characteristic constant T_2 is measured by fitting the curve 3.20 to a set of data points obtained by initializing the system in the state $\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$. During free evolution, $R_x(\pi)$ pulses are applied at $t/2$ and t, implementing the spin echo technique to reduce additional external effects on the qubit, such as coupling, magnetic field inhomogeneity, and noise. Due to the spatial distribution of molecules in the sample within the magnetic field, any inhomogeneity in the field causes a difference in the precession frequency around the field axis. These differences accumulate phase variations that reduce the coherence of the state.

The effect of the $R_x(\pi)$ pulses is, in approximation, to cancel the errors introduced by these factors on average, by flipping the direction of the longitudinal evolution.

The Rabi relaxation time, T_2^*, is further defined, corresponding to the same type of transverse relaxation as T_2, but in the absence of spin echo pulses. It represents the relaxation of the $\langle S_{xy}\rangle$ component when it is also affected by factors such as those mentioned previously, and can be considered the limit of system coherence in the worst-case scenario.

3.3.3 Time requirements

According to DiVincenzo, relaxation times should be 10^4 to 10^5 times longer than quantum gate operation times. In the two-qubit sample system used in the Gemini Lab quantum computer, we measure $T_1 \approx 11\,\mathrm{s}$ and $T_2 \approx 4\,\mathrm{s}$. The J-coupling interaction used to implement two-qubit gates imposes minimum gate durations on the order of milliseconds. This requirement is therefore satisfied.

However, reducing the duration of the designed quantum gates remains an important objective in pulse sequence optimization.

3.4 DiVincenzo 4: Universal set of Quantum Gates

To use the two-level system that defines each qubit, it is necessary to be able to transform the system from any initial state ψ_0 to any target state ψ_1. Formally, quantum gate universality is the ability to achieve any desired evolution using a finite set of quantum gates.

There are several possible universal sets of quantum gates. The most common choice consists of single-qubit rotations, together with any non-trivial two-qubit gate, i.e., a gate that cannot be decomposed into independent single-qubit rotations. Such gates typically rely on an intrinsic coupling interaction between two qubits in the system.

In NMR, as a consequence of the system model given by Eq. 3.12 and described in Section 3.1, any single-qubit rotation can be implemented through the control Hamiltonian term, since it has components along I_x and I_y. Rotations about the $\hat{x}$ and $\hat{y}$ axes are achieved using pulses that differ only in phase, while rotations about the $\hat{z}$ axis can be implemented either through free precession around the constant field direction $\vec{B}_0$, or, equivalently, by decomposing them into rotations about transverse axes.

Using Euler angle decomposition, an arbitrary single-qubit rotation can be expressed as a sequence of rotations about two orthogonal axes in the transverse plane (e.g., $\hat{x}$ and $\hat{y}$), such as $R_x(\alpha)R_y(\beta)R_x(\gamma)$, which reproduces any effective rotation including those about $\hat{z}$. The J-coupling term is used for the implementation of multi-qubit gates.

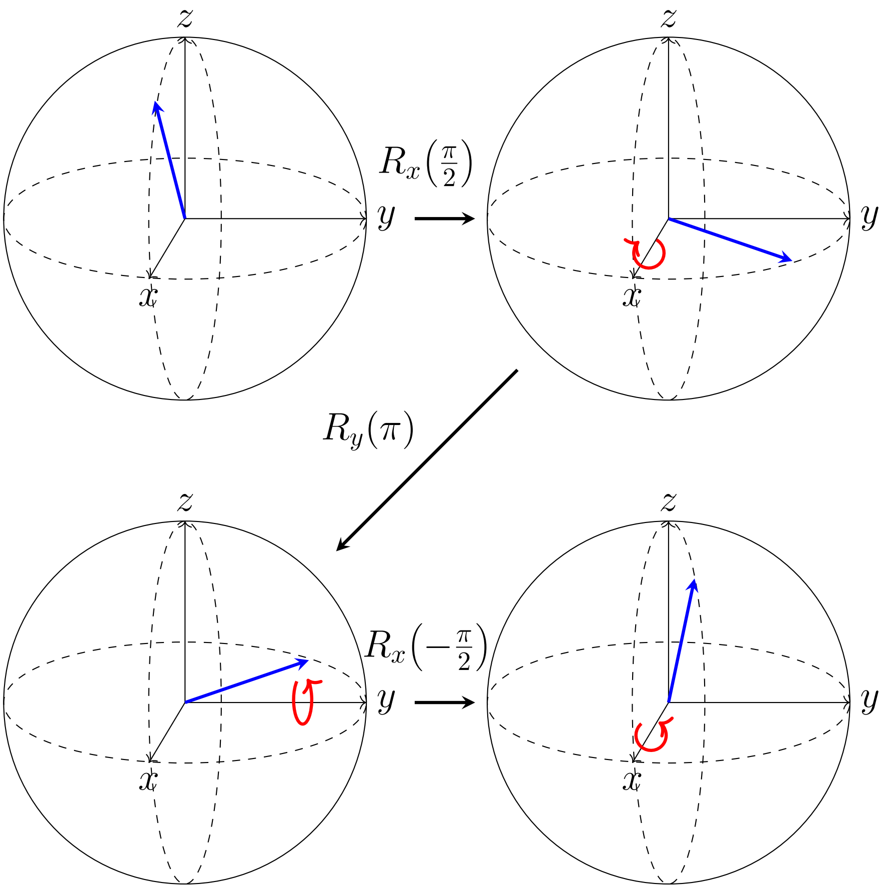

Figure 3.8: An arbitrary single-qubit rotation can be decomposed into three successive rotations about the x and y axes. Here, the rotation $R_z(\pi)$ is implemented using the Euler decomposition $R_z(\pi) = R_x(\frac{\pi}{2})\,R_y(\pi)R_x(-\frac{\pi}{2})$, illustrating the evolution of the Bloch vector after each elementary rotation.

An important point to consider is that the implementation of some of these gates requires spin-selective excitation. As mentioned previously, in heteronuclear systems the different nuclei have significantly distinct Larmor precession frequencies. This makes each qubit easily identifiable and individually addressable in the frequency domain.

In homonuclear systems, the Larmor frequencies of different qubits are very close to each other, making them difficult to distinguish in the frequency domain. Nevertheless, small variations in these frequencies arise due to the electronic environment and the molecular geometry, known as chemical shifts. When these differences are sufficiently large, they can be exploited to design selective pulses capable of acting on specific qubits.

When a clear separation cannot be achieved through chemical shifts, the multi-qubit system must be manipulated collectively, taking into account the joint behavior of the entire system.

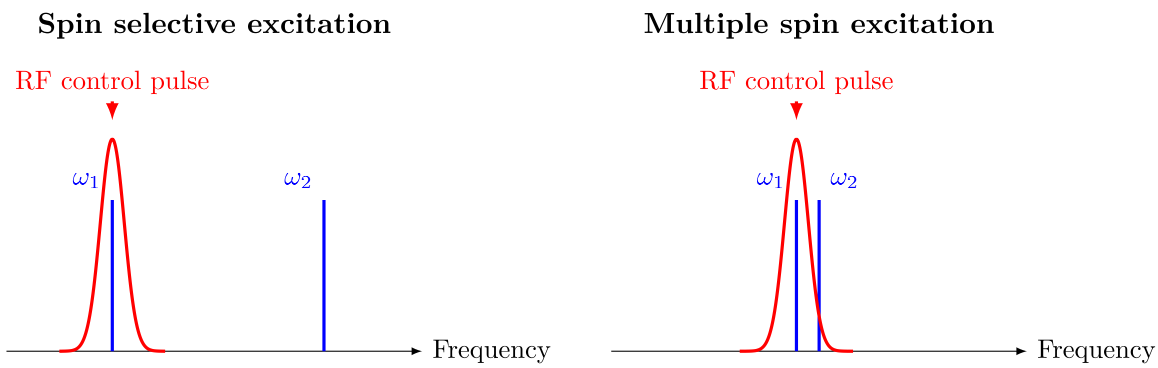

Figure 3.9: Selective and collective excitation in NMR systems. A radio frequency control pulse excites only the resonant nuclei when the Larmor frequencies are separated by a sufficiently large margin (left). When the frequencies are close to each other, the pulse overlaps nearby resonances causing partial excitation. This effect occurs in homonuclear systems when the chemical shift is too small (right).

3.5 DiVincenzo 5: Individual measurement

3.5.1 Free Induction Decay

The measurement of each qubit state is not performed by directly measuring individual nuclear spins. The quantum system is an ensemble of approximately 10^{23} molecules, each containing as many relevant nuclear spins as there are nuclei used as qubits. The measured quantity is, instead, the macroscopic magnetic moment of the system, $\hat{M} = (M_x, M_y, M_z)$, whose components are proportional to the statistical average of the corresponding spin components $\langle S_x \rangle, \langle S_y \rangle, \langle S_z \rangle$, i.e., the nuclear spin angular momentum.

Thus, when referring to the state of a qubit, one is not describing the spin state of a single molecule, but rather the ensemble average $\hat{M}$.

When an RF field is applied at the same frequency as the intrinsic precession frequency of the qubit spin, ω_0, i.e., its Larmor frequency, the energy of the field matches the energy difference between the qubit levels. During this resonance between the field and the qubit, the latter can absorb energy and transition from the lower-energy state to the excited state, resulting in a rotation of the M_z component toward the negative direction of the $\hat{z}$ axis, or, via stimulated emission, undergo the reverse process.

Simultaneously, the transverse component of the net magnetic moment precesses around the axis of the static magnetic field $\vec{B}_0$ at frequency ω_0, as it reflects the ensemble average behavior of each nuclear spin as described in Section 3.1. The time evolution of the M_x and M_y components, also including decoherence effects, can be expressed as

$$\begin{aligned} M_x(t) &= |M_{xy}(t_0)| \cos(2\pi\omega_0 t + \phi) e^{-t/T_2}, \\ M_y(t) &= |M_{xy}(t_0)| \sin(2\pi\omega_0 t + \phi) e^{-t/T_2}. \end{aligned} \tag{3.21}$$

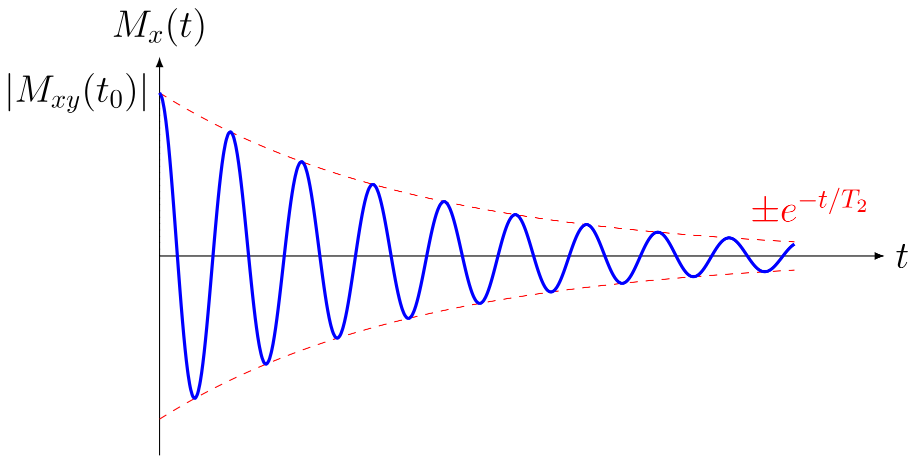

Figure 3.10: Free induction decay (FID) generated by the transverse magnetization after RF excitation. The detected signal oscillates at the Larmor frequency ω_0 while its amplitude decays exponentially with the transverse relaxation time T_2, as described by the exponentially damped transverse magnetization.

The evolution of the transverse component of the magnetic moment generates an electromagnetic field that is detected by the system's coils, known as the Free Induction Decay (FID), which contains the information required to reconstruct the initial quantum state.

The frequency-domain counterpart of the FID signal, obtained via a Fourier transform, is referred to as a Complex Lorentzian. The magnitudes of these frequency-domain signals are proportional to the values of the M_x and M_y components prior to the onset of relaxation. To measure M_z, the quantum state is rotated so that it is mapped onto the M_x axis, after which it is measured in the same manner.

3.5.2 Quantum state tomography

Quantum state tomography is the process of reconstructing a quantum state utilizing information from multiple diferent measurements of identical states. These measurements must use operators that form a complete basis on the Hilbert space of the system, so that the full density matrix can be reconstructed.

Realizing quantum tomography on the state of a single qubit is simple. The density matrix can be fully reconstructed using the Pauli operators, which translate to measuring the Bloch vector, $\vec{r}$, in the directions $\hat{x}, \hat{y}, \hat{z}$. The process of measurement causes the collapse of the quantum state onto one of the possible eigenstates. This is a probabilistic outcome, so multiple measurements of the same observable can be conducted to recover the initial component of the Bloch vector in its direction.

$$\rho = \frac{1}{2}\left(I + \vec{r}\cdot\vec{\sigma}\right) = \frac{1}{2}\begin{bmatrix} 1 + r_z & r_x - i r_y \\ r_x + i r_y & 1 - r_z \end{bmatrix} \tag{3.22}$$

However, quantum tomography of multi-qubit systems isn't that simple. As the number of qubits increases, the number of observables required to reconstruct the density matrix also increases, and finding measurement basis that provide information efficiently can be challenging. In addition, the possibility of quantum entanglement existing between two qubits prevents the reconstruction of the complete quantum state of a composite system from separate qubit measurements.

In that case, the Complex Lorentzian resulting from the Fourier transform of the Free Induction Decay signal previously mentioned must be studied.

$$\begin{aligned} L_{M_x} &= \frac{|a_l|[\cos(\phi)\lambda + \sin(\phi)(\omega - \omega_0)]}{\lambda^2 + (\omega - \omega_0)^2} \\ L_{M_y} &= \frac{|a_l|[\sin(\phi)\lambda + \cos(\phi)(\omega - \omega_0)]}{\lambda^2 + (\omega - \omega_0)^2} \end{aligned} \tag{3.23}$$

4 Radio Frequency Pulse Design

In this chapter, I present the concepts and methods used for the design of radio-frequency control pulses. Pulse sequences may be designed directly by exploiting the control Hamiltonian from equation 3.12 and selecting appropriate values for its parameters, such as pulse amplitude, phase, frequency, and duration. Alternatively, pulse design can be formulated as an optimal-control problem, in which these parameters are determined through numerical optimization. I will first outline the general principles behind this approach and then discuss in greater detail the most widely used methodology, Gradient Ascent Pulse Engineering (GRAPE), which was also employed in this project.

4.1 Analytical design: Hard pulses

4.1.1 Single-qubit operations

Analyzing the control Hamiltonian to select appropriate pulse parameters is a straightforward approach to pulse design. Simple operations such as single-spin rotations can be implemented using rectangular pulses, commonly referred to as **hard pulses**. Provided that the Larmor frequencies of the qubits are sufficiently well separated and that the pulse duration t is short enough so that both the first two terms of equation 3.12 can be ignored, the dynamical evolution following the remainder of the control Hamiltonian is described as

$$U(t) = e^{-iHt} = e^{-i2\pi\omega_1^i[\cos(\phi_{rf})I_x^i + \sin(\phi_{rf})I_y^i]t}. \tag{4.1}$$

This unitary operator represents a rotation around the axis $\vec{\sigma} = \cos(\phi_{rf})\hat{x} + \sin(\phi_{rf})\hat{y}$ in the Bloch sphere, by an angle of $\theta = 2\pi\omega_1^i t$. As an example, consider the sequence of hard pulses shown in Figure 4.1. The three pulses implement an Euler angle decomposition through successive rotations around different axes in the transverse plane. The phase of each pulse determines the rotation axis, while the pulse duration controls the rotation angle according to equation 4.1.

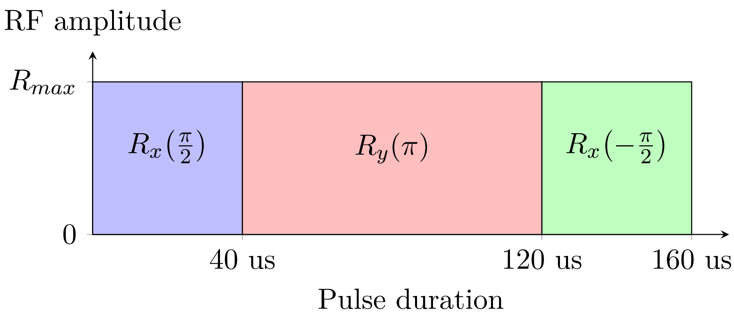

Figure 4.1: The example Euler angle rotation shown previously in Figure 3.8 can be implemented using only maximum amplitude hard pulses. The $R_x(\pi/2)$ rotation is a 40 µs pulse with phase 0°, the $R_y(\pi)$ rotation is an 80 µs pulse with phase 90°, and the final $R_x(-\pi/2)$ rotation is a 40 µs pulse with phase 180°.

In this way, any arbitrary rotation of a single-spin state can be achieved using a finite number of hard pulses, by adjusting their phase and duration. This establishes hard pulses as a powerful tool for implementing single-spin rotations due to their short duration and high fidelity in the appropriate regime. As a result, they are often used to implement single-qubit rotation gates about the $\hat{x}$, $\hat{y}$, and $\hat{z}$ axes, as well as the Hadamard gate, which can be represented as an effective π-rotation about the axis $\frac{1}{\sqrt{2}}(\hat{x}+\hat{z})$.

4.1.2 Multi-qubit operations

For multi-qubit operations, the J-coupling term cannot be neglected as this inter-qubit interaction is used to mediate the transfer of information between the qubits involved. Since it is an always-on interaction, its effect on the system's evolution must be taken into account over the entire duration of the control pulse. For a two-qubit system, the free evolution under just the J-coupling interaction, in the absence of a control pulse, is given by

$$U(t) = e^{-iHt} = e^{-i2\pi J_{12} I_z^1 I_z^2 t}. \tag{4.2}$$

This represents a rotation by an angle of $\theta = 2\pi J_{12} t$, around the axis given by the operator $I_z^1 I_z^2$. This does not translate to a rotation around a single spatial axis like $\hat{x}$, $\hat{y}$, or $\hat{z}$, but rather to a conditional phase evolution that depends on the state of the spins. Expanding the operator in terms of the Pauli operators for each spin yields

$$\begin{aligned} I_z^1 \otimes I_z^2 &= \frac{1}{2}\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \otimes I_z^2 \\ &= \frac{1}{2}\left(|0\rangle\langle 0| \otimes I_z^2 + |1\rangle\langle 1| \otimes -I_z^2\right). \end{aligned} \tag{4.3}$$

The result is that the rotation axis is $\hat{z}$ if the first qubit is in the state $|0\rangle$, and $-\hat{z}$ if it is in the state $|1\rangle$. The same analysis can be performed for the second qubit.

This can be used for the analytical construction of multi-qubit gates such as the Controlled-NOT gate. This gate applies a NOT operation (an X gate) on the target qubit only when the control qubit is in the state $|1\rangle$. Geometrically, this corresponds to a conditional π-rotation of the target qubit about the $\hat{x}$-axis on its Bloch sphere, conditioned on the state of the control qubit.

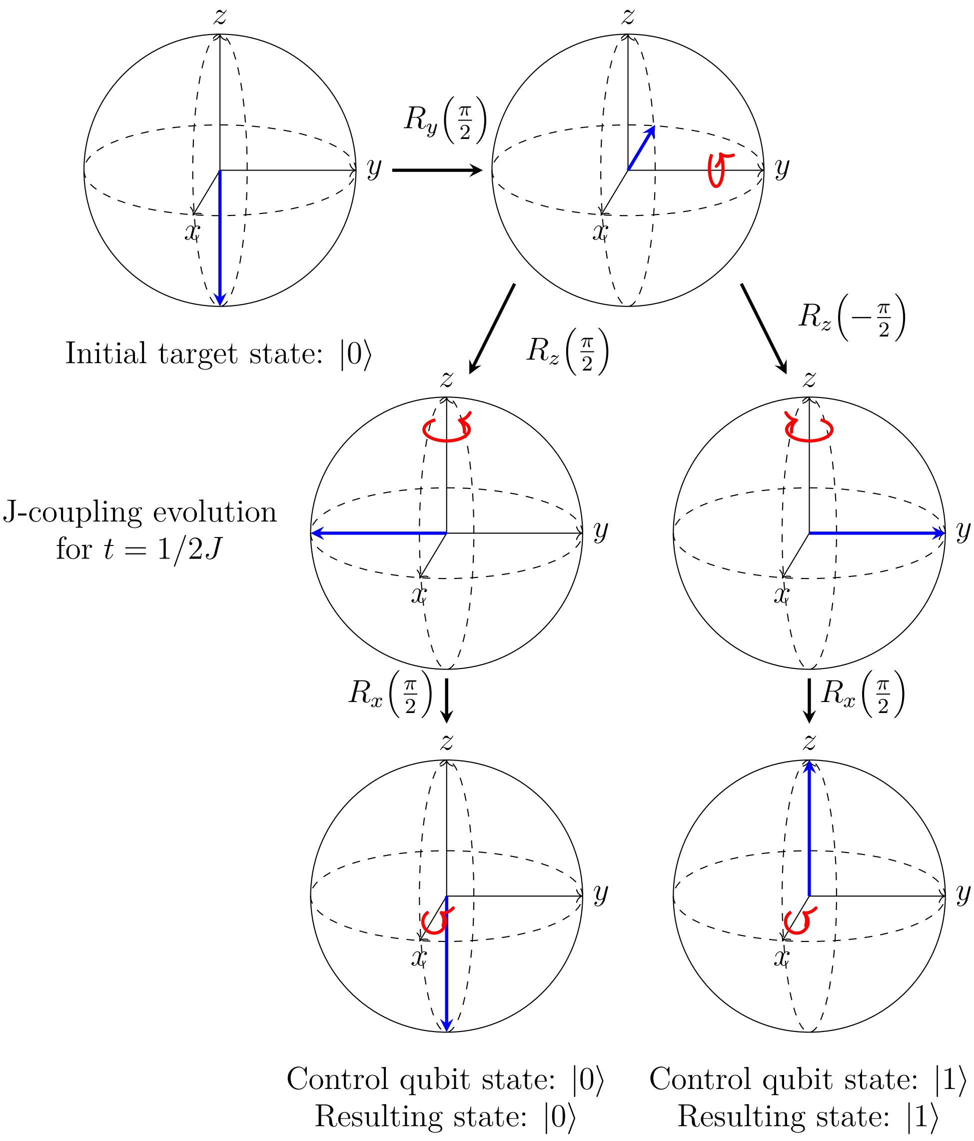

Figure 4.2: A CNOT gate can be implemented through a sequence of hard pulses making use of the intrinsic J-coupling evolution. The target qubit is rotated to the transverse plane, where it evolves under the J-coupling interaction for a time $t = 1/2J$, which corresponds to a conditional $\pi/2$ rotation about the $\hat{z}$ or $-\hat{z}$ axis.

However, deriving pulse sequences analytically for more complex operations or for systems with more than two qubits quickly becomes unmanageable. This is mainly due to the always-on nature of the J-coupling interaction, which cannot be switched off or modulated in strength. In addition, practical pulse design must account for the fact that a single hard pulse can unintentionally excite multiple spins simultaneously, making selective addressing and decoupling during gate implementation non-trivial.

4.2 Numerical optimization

To circumvent the problems of analytical pulse design, especially in the presence of the aforementioned issues, numerical optimization methods are employed to systematically search for pulse parameters that achieve the desired quantum evolution with high fidelity.

Section 2.1.2 established that an ideal quantum gate is described by a unitary operation. Consequently, the goal of a pulse optimization algorithm is to find the pulse parameters that maximize the similarity between the evolution generated by the pulse and the target unitary operation.

4.2.1 Defining a cost function

Every numerical optimization algorithm begins with the definition of a cost function. The objective is to determine a set of parameters that either maximize or minimize this function.

In the present work, the cost function is defined as the **fidelity** of the quantum gate. It is computed by measuring the overlap between the target unitary operation and the unitary operation generated by the optimized pulse, making use of the Hilbert-Schmidt inner product, and normalized afterward:

$$F = \frac{\left|\mathrm{Tr}(U_{target}^{\dagger} U_{optimized})\right|^2}{d^2}. \tag{4.4}$$

It is important to distinguish between theoretical fidelity and experimentally achieved fidelity. The fidelity defined above is based on the assumption that the system dynamics behave exactly as described by the idealized Hamiltonian used during optimization (equation 3.12). However, this Hamiltonian is a product of multiple approximations and further experimental phenomena can lead to discrepancies between the optimized and implemented unitary operations.

The cost function can incorporate further objectives beyond fidelity, such as robustness and smoothness constraints. **Robustness** refers to the insensitivity of the optimized control pulse to small perturbations in system parameters, such as fluctuations in the magnetic field strength. This is incorporated into the cost function by evaluating the fidelity of the optimized pulse for a range of slightly perturbed Hamiltonians, and taking an average of the results.

Smoothness characterizes the temporal variation of the control amplitudes. Smooth control pulse profiles are encouraged because abrupt changes may not be faithfully reproduced by the experimental hardware due to finite bandwidth and transient rise-time effects, leading to discrepancies between the designed and implemented pulses.

In a similar vein, the optimized pulse can also be constrained to follow a **gaussian-square envelope**, which is a common pulse shape in NMR experiments. A smooth and physically realizable rise and fall time can be beneficial when considering the equipment's physical limitations, as well as that the optimized pulse will ideally be incorporated into a larger pulse sequence to form an algorithm.

In this work, the pulse envelope was implemented by performing a convolution of the pulse parameters with the desired envelope function, during the optimization process. However, it was later found through experimentation that this constraint was not necessary, as the experimental equipment was able to apply the optimized pulses faithfully.

4.2.2 Optimization parameters

Having characterized the cost function, the next step is the definition of the optimization parameters that will give shape to the optimized pulse. In Section 3.1.3, the control Hamiltonian was defined in terms of three parameters: the Rabi oscillation frequency ω_1, the RF field phase ϕ_{rf}, and the reference frame frequency ω_{rf}, in equation 3.12. In addition, the pulse duration t is also a parameter that can be controlled.

Under the lens of numerical optimization, these parameters are interpreted as the degrees of freedom that can be adjusted to maximize the cost function. The optimization parameters are therefore:

1. **Detuning**: δ_{rf}

   By varying the reference frame frequency ω_{rf}, the system can be driven off-resonance from the Larmor frequency ω_0 of the qubit, introducing a detuning term $\delta_{rf} = \omega_{rf} - \omega_0$. In the rotating frame, this results in an additional effective z-component in the Hamiltonian, which modifies the rotation axis of the control field and, consequently, the efficiency and direction of the implemented operation.

   In homonuclear systems, this variation can be useful for achieving approximate single-qubit addressing and selective excitation. However, in heteronuclear systems such as the one used in this work, each qubit has a discernible Larmor frequency, making individual addressing straightforward.

2. **Phase**: ϕ_{rf}

   As a result of the description of the system by equation 3.12, the phase of the RF pulse dictates the axis of rotation of the unitary operation. A phase of $\phi_{rf} = 0$ defines a rotation about the $\hat{x}$ axis, whereas a phase of $\phi_{rf} = \pi/2$ defines a rotation about the $\hat{y}$ axis.

3. **RF field amplitude percentage**: R

   The amplitude of the RF field is defined relative to the maximum calibrated Rabi frequency of each qubit, corresponding to the maximum achievable RF amplitude. It is expressed as:

$$R = \frac{\omega_1}{\omega_1^{\max}} = \frac{B_1}{B_1^{\max}} \times 100\% \tag{4.5}$$

   The amplitude is directly related to the effective speed of the implemented unitary operation. Higher amplitudes are generally preferred, as they reduce the pulse duration required to perform the desired operation, thereby mitigating decoherence effects during long control sequences.

4. **Pulse duration**: t

   The pulse duration represents the time required to implement the target unitary operation. As shorter duration pulses are always preferred to avoid decoherence in longer pulse sequences, the optimization process should aim to minimize this parameter whenever possible. Nevertheless, the minimum achievable pulse length is limited by the physical dynamics of the system, particularly the time required for inter-qubit interactions through J-coupling evolution.

The total pulse duration is discretized into N time steps, within which the control parameters are held constant. Each optimized pulse therefore consists of an ordered sequence of hard pulses U_k, whose parameters are adjusted such that the successive application of the corresponding elementary unitary operations approximates the target quantum gate.

$$\begin{aligned} U_k &= e^{-iH(R^k,\omega_{rf}^k,\phi_{rf}^k)t^k} \\ U_{\text{optimized}} &= U_N U_{N-1} \cdots U_2 U_1 \end{aligned} \tag{4.6}$$

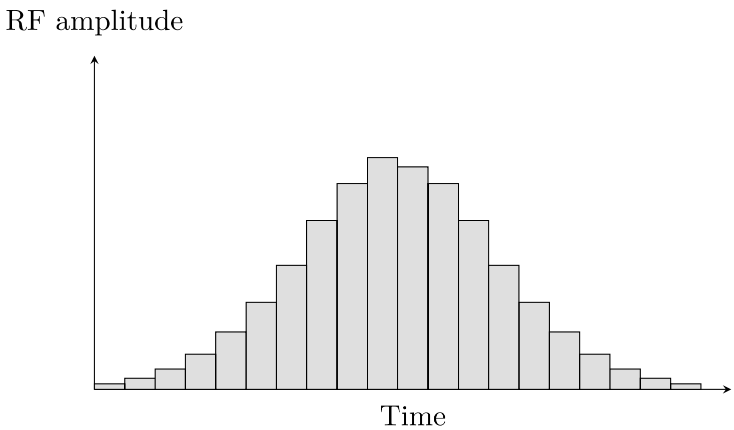

Figure 4.3: Pulse discretization. The continuous pulse is divided into N time steps. Within each time step, the control parameters are held constant, effectively resulting in a sequence of hard pulses, which are optimized to approximate the target unitary operation.

4.2.3 Gradient Ascent Pulse Engineering

To identify an optimal solution of a cost function, a common strategy is the use of gradient ascent and gradient descent algorithms. These methods compute the gradient of the cost function in parameter space and update the parameters in the direction of steepest increase or decrease of the cost function every iteration. The optimization is conducted iteratively until predefined convergence criteria are met, typically based on the magnitude of the gradient or the variation of the cost function between iterations.

An ordinary gradient based algorithm will obey the following workflow in Figure 4.4.

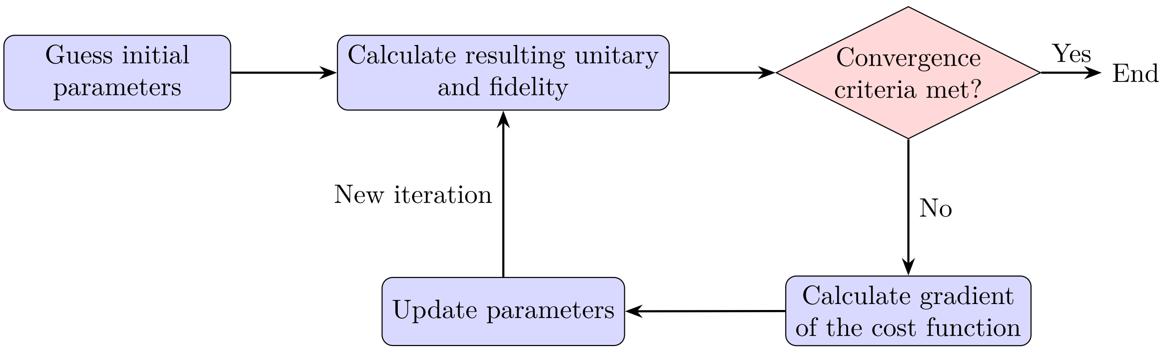

Figure 4.4: The algorithm begins with an initial guess of the control parameters, either arbitrary or based on prior knowledge of the system. The resulting unitary operation and its respective fidelity are computed as shown in equation 4.4. If the convergence criteria are not met, the gradient of the cost function with respect to each of the control parameters is calculated after which the parameters are updated in the direction of the gradient. This process is repeated iteratively until the convergence criteria are satisfied.

As the system size increases, so does the complexity of its dynamics, as well as the number of parameters required to accurately describe them. Consequently, the associated computational cost of numerical optimization quickly becomes significant. In this context, one of the most widely used techniques for pulse design in Nuclear Magnetic Resonance is Gradient Ascent Pulse Engineering. Introduced by Khaneja et al., GRAPE provides an efficient framework for solving a range of optimal control problems by enabling the systematic and computationally manageable optimization of control pulses [26]. Among the challenges it addresses is the design of control sequences that implement a desired unitary propagator, which is also the central objective of this work. The following description of the GRAPE algorithm is based on Khaneja et al.[26], in particular Section 2.4 of their work.

To efficiently explore the parameter space, a GRAPE algorithm begins by reparameterizing the control field from its polar representation, defined by an amplitude and phase, to a cartesian representation with independent control amplitudes along the orthogonal directions $\hat{x}$ and $\hat{y}$, denoted by R_x and R_y. This parametrization ensures that the optimization variables have similar numerical behavior and avoids complications associated with the periodic nature of the phase parameter.

$$R_x = R\cos\phi_{rf} \qquad R_y = R\sin\phi_{rf} \tag{4.7}$$

$$H_k = \sum_{i=1}^{n} 2\pi\hbar \left(R_x^i \omega_1^{\max} I_x + R_y^i \omega_1^{\max} I_y \right) \tag{4.8}$$

Furthermore, for a radio frequency pulse divided into N time steps of duration Δt, the control Hamiltonian at each time step j, where H_0 is the internal Hamiltonian of the system, H_k are the control Hamiltonians (in this case just the one in equation 4.8), and $u_k(j)$ are the control amplitudes to be optimized, is given by

$$H_{\text{control}}(j) = H_0 + \sum_k u_k(j) H_k. \tag{4.9}$$

Now, what distinguishes GRAPE from other optimization frameworks is the following step: at the beginning of each optimization iteration, the cumulative unitary evolution up to each time step and the corresponding backward-propagated target evolution are computed. The forward propagation is usually denoted by X_j and the backward propagation by P_j.

$$\begin{aligned} P_j &= U_{j+1}^\dagger U_{j+2}^\dagger \cdots U_N^\dagger U_{target} \\ X_j &= U_j U_{j-1} \cdots U_1 \end{aligned} \tag{4.10}$$

These forward and backward propagators are then used to compute the gradient of the fidelity with respect to each control amplitude $u_k(j)$, which quantifies how a small change in each of the parameters affects the final quantum gate's fidelity, and is given by

$$\frac{\delta F}{\delta u_k(j)} = -2\mathrm{Re}\langle P_j | i\Delta t H_k X_j \rangle \langle X_j | P_j \rangle. \tag{4.11}$$

Each control parameter $u_k(j)$ is then updated in the direction of the gradient scaled by a factor α, known as step size or learning rate. This factor must be fine-tuned appropriately because a value that is too small leads to slow convergence and unnecessary computational expense, whereas a value that is too large may cause the optimization to overshoot and fail to converge to a local optimum. This will be discussed further in the following Section 4.2.4.

$$u_k^{(m+1)}(j) = u_k^{(m)}(j) + \alpha \frac{\delta F}{\delta u_k(j)} \tag{4.12}$$

After updating the control amplitudes, a new pulse sequence is obtained and the corresponding unitary operation and fidelity are recomputed. This process is repeated iteratively until a predefined convergence criterion is satisfied, typically a target fidelity or a maximum number of iterations. The result is a set of control amplitudes $R_x(j)$ and $R_y(j)$ defining a pulse sequence that maximizes the overlap between its unitary operator and the target quantum gate.

The key insight that distinguishes GRAPE from other optimization frameworks is the reduced number of matrix exponential computations. By computing the forward and backward propagations only once per iteration, rather than once per parameter update, the computational cost is significantly reduced, enabling efficient optimization even for large-scale control problems.

4.2.4 Function Optimizers

The optimization step of the algorithm, in which the parameters are updated based on the gradient scaled by a factor α, is particularly sensitive and must be carefully tuned. Rather than manually adjusting the learning rate, the use of more advanced optimization routines can lead to faster convergence and improved performance.

One such optimization algorithm is **Limited-memory BFGS (BFGS)**, which is widely used in machine learning and optimal control for parameter estimation. It approximates second-order information by building an estimate of the inverse Hessian matrix, enabling efficient navigation of the parameter space toward a local minimum. The method is memory-efficient, as it stores only a limited number of vectors instead of the full Hessian matrix, making it well suited for problems with a large number of optimization parameters.

In this work, L-BFGS was the sole optimization algorithm used for generating control pulses using GRAPE. However, one of its limitations is its susceptibility to convergence toward suboptimal local minima. When the cost function is represented by a complex landscape with multiple local minima, the optimizer may become trapped in non-optimal solutions, making it difficult to obtain high-fidelity results. To prevent this, a good initial guess that lies in the vicinity of an optimal solution is often required. Alternatively, more robust optimization algorithms can be used, which are less sensitive to local minima but typically require higher computational cost and slower convergence.

An alternative algorithm is **Adaptive Moment Estimation (ADAM)**, a stochastic gradient descent method that combines momentum-based updates with adaptive learning rates. Each update is computed using an exponentially weighted average of past gradients and past squared gradients, which improves stability and reduces sensitivity to the choice of step size. The effective learning rate is automatically adapted at each iteration based on these moment estimates, leading to more robust convergence in complex optimization landscapes.

5 Implementation of the SWAP test

In this chapter, I present the implementation of a two-qubit destructive SWAP test, introduced by García-Escartín et al. [27]. Unlike the conventional SWAP test, this protocol does not require an ancillary qubit, at the cost of requiring the measurement of the target quantum states. I also show how the scalar product between the two quantum states can be extracted from the measurement outcomes of the destructive SWAP test. Finally, I describe the design and successive optimization of a control pulse sequence that implements this operation on the NMR quantum computer used in this work.

5.1 The SWAP test

A conventional SWAP test requires three qubits, one of which serves as an ancillary qubit. The circuit begins with a Hadamard gate applied to the ancilla, followed by a controlled-SWAP (Fredkin) gate acting on the two target qubits and controlled by the state of the ancilla. A second Hadamard gate is then applied to the ancillary qubit before measurement.

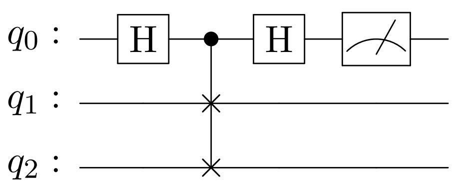

Figure 5.1: Conventional SWAP test

$$
\begin{aligned}
|0\rangle|\psi\rangle|\phi\rangle &\xrightarrow{\text{Hadamard}} \frac{|0\rangle + |1\rangle}{\sqrt{2}}|\psi\rangle|\phi\rangle \\
&\xrightarrow{\text{CSWAP}} \frac{|0\rangle|\psi\rangle|\phi\rangle + |1\rangle|\phi\rangle|\psi\rangle}{\sqrt{2}} \\
&\xrightarrow{\text{Hadamard}} \frac{|0\rangle[|\psi\rangle|\phi\rangle + |\phi\rangle|\psi\rangle] + |1\rangle[|\psi\rangle|\phi\rangle - |\phi\rangle|\psi\rangle]}{2}
\end{aligned}
\tag{5.1}
$$

If the two quantum states are identical, measuring the ancillary qubit always yields the state $|0\rangle$, since the antisymmetric component of the joint state vanishes: $|\psi\rangle|\phi\rangle - |\phi\rangle|\psi\rangle = 0$. More generally, the probability of obtaining the outcome $|0\rangle$ is

$$\begin{aligned} P(|0\rangle) &= \frac{1}{4} \parallel |\psi\rangle|\phi\rangle + |\phi\rangle|\psi\rangle \parallel^2 \\ &= \frac{1 + |\langle\psi|\phi\rangle|^2}{2}. \end{aligned} \tag{5.2}$$

Conversely, if the two states are orthogonal, then $|\langle\psi|\phi\rangle| = 0$, and the ancilla is measured in the states $|0\rangle$ and $|1\rangle$ with equal probability. Repeating the experiment multiple times therefore yields each outcome with probability 1/2.

Motivated by the Hong–Ou–Mandel effect in quantum optics, which is used to compare the states of two photons, García-Escartín et al. proposed in 2013 a protocol to perform a SWAP test without the need for an ancillary qubit [27]. They demonstrate that this protocol is equivalent to the conventional SWAP test by decomposing it into simpler, equivalent circuits in which the ancillary qubit is removed, and the protocol is reformulated in terms of direct measurements on the two target qubits. They further show that the method can be generalized to an arbitrary number of qubits. The circuit in Figure 5.2 illustrates their two-qubit implementation.

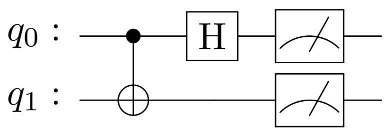

Figure 5.2: Destructive SWAP test

This circuit is effectively a measurement in the Bell basis and yields the same measurement statistics as the conventional SWAP test by measuring the two target qubits. In the destructive SWAP test, the measurement outcome $|11\rangle$ corresponds to a projection onto the antisymmetric Bell state, $|\Psi^-\rangle$, whose amplitude is proportional to the antisymmetric component of the joint input state. Consequently, if the two input states are identical, this antisymmetric component vanishes and the probability of measuring $|11\rangle$ is zero.

Consider the initial states $|\psi\rangle$ and $|\phi\rangle$:

$$|\psi\rangle = a|0\rangle + b|1\rangle, \qquad |\phi\rangle = c|0\rangle + d|1\rangle. \tag{5.3}$$

Applying the destructive SWAP test yields

$$\begin{aligned}
ac|00\rangle + ad|01\rangle + bc|10\rangle + bd|11\rangle &\xrightarrow{\text{CNOT}} ac|00\rangle + ad|01\rangle + bc|11\rangle + bd|10\rangle \\
&\xrightarrow{\text{Hadamard}} \frac{1}{\sqrt{2}}\big[ac|00\rangle + ac|10\rangle + ad|01\rangle + ad|11\rangle \\
&\qquad + bc|01\rangle - bc|11\rangle + bd|00\rangle - bd|10\rangle\big].
\end{aligned} \tag{5.4}$$

From this expression, the probability of measuring $|11\rangle$ is given by

$$\begin{aligned}
P(|11\rangle) &= \frac{|ad - bc|^2}{2} \\
&= \frac{(ad - bc)(ad - bc)^*}{2} \\
&= \frac{|a|^2|d|^2 - adb^*c^* - bca^*d^* + |b|^2|c|^2}{2} \\
&= \frac{1 - |\langle\psi|\phi\rangle|^2}{2}.
\end{aligned} \tag{5.5}$$

The complementary outcome to measuring the state $|11\rangle$, corresponding to the measurement of any of the remaining three computational basis states, yields the same probability as in Eq. 5.2

$$\begin{aligned}
P_{\text{pass}} &= 1 - P(|11\rangle) \\
&= P(|00\rangle) + P(|01\rangle) + P(|10\rangle) \\
&= \frac{1 + |\langle\psi|\phi\rangle|^2}{2}.
\end{aligned} \tag{5.6}$$

The equivalence between the conventional SWAP test and the destructive SWAP test is thus established. Analogously to measuring the ancilla qubit in the state $|0\rangle$ in the conventional SWAP test, a single measurement outcome other than $|11\rangle$ is not sufficient to determine whether the input states are identical, as the protocol relies on probabilistic measurement outcomes. Consequently, the overlap between the input states must be inferred from measurement statistics accumulated over many repetitions. In an NMR quantum computer, however, this requirement is naturally satisfied, since measurements are performed over an ensemble of identically prepared molecules and directly yield expectation values encoded in the measured density matrix.

From these statistical results, the scalar product between the two tested states can be extracted. This is required for the evaluation of Bargmann invariants, which are defined as products of inner products between sequences of quantum states. The destructive SWAP test therefore provides direct access to the overlaps needed to experimentally determine these geometric quantities.

5.2 Pulse design

5.2.1 GRAPE optimization

Using the algorithm described in Section 4.2.3, a search was performed for a single control pulse implementing the complete unitary operation of the destructive SWAP test. The control amplitudes were initialized to a near-zero value (0.0001%), the time step was fixed at 4 μs, and the optimization included robustness against 5% RF field amplitude variations together with a smoothness penalty of weight 5×10^{-3}.

A time sweep was then carried out for pulse durations ranging from 1 ms to 5 ms. Candidate pulses were selected for experimental implementation according to three criteria: high theoretical fidelity, short pulse duration, and a physically reasonable pulse profile. The theoretical fidelity obtained as a function of pulse duration is shown in Figure 5.3.

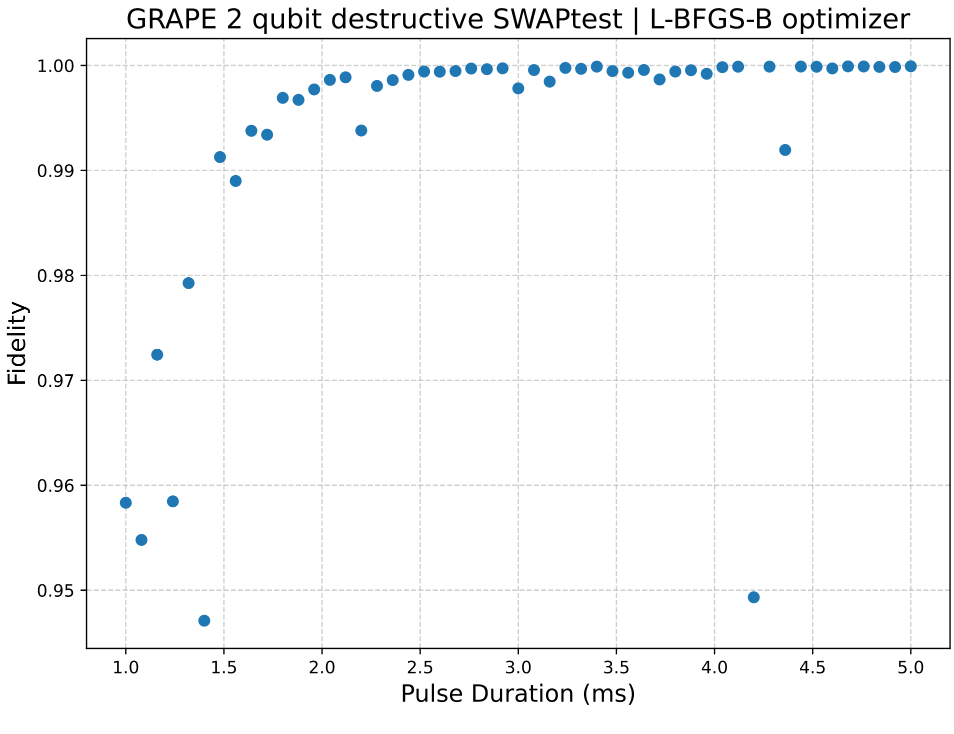

Figure 5.3: GRAPE-implemented destructive SWAP test gate theoretical fidelity for different pulse durations. Each data point represents the best theoretical fidelity obtained for a given pulse duration.

The experimental fidelity was evaluated by applying the control pulse to each of the computational basis states, $|00\rangle$, $|01\rangle$, $|10\rangle$, and $|11\rangle$. The experimentally reconstructed output state for each input was then compared with the corresponding ideal target state using the Hilbert-Schmidt inner product. The reported experimental fidelity is the average of these four state fidelities, given by

$$F_{\text{experimental}} = \frac{1}{N} \sum_{i=1}^{N} \mathrm{Tr}\Big(\rho^{(i)}_{\text{experimental}} \rho^{(i)}_{\text{target}}\Big) , \tag{5.7}$$

where $N = 4$ is the number of computational basis states.

The best-performing candidates, with durations of 1.8 ms and 2.12 ms, achieved theoretical fidelities exceeding 99%. Experimentally, however, both pulses reached fidelities of only approximately 70%, revealing a significant discrepancy between the idealized optimization model and the physical implementation. The corresponding pulse profiles are shown in Figures 5.4 and 5.5.

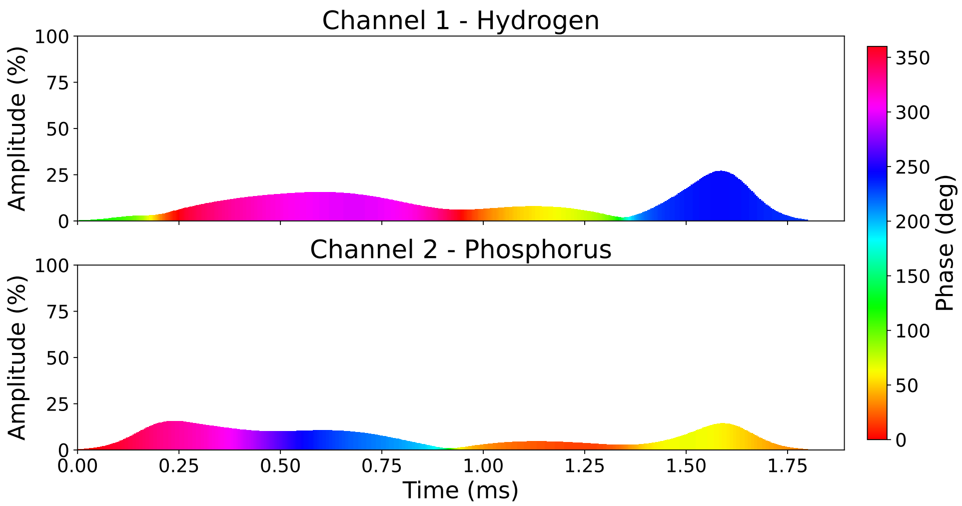

Figure 5.4: Pulse profile for the SWAP test optimization with a duration of 1.8 ms. Experimental fidelity of 69.2544%.

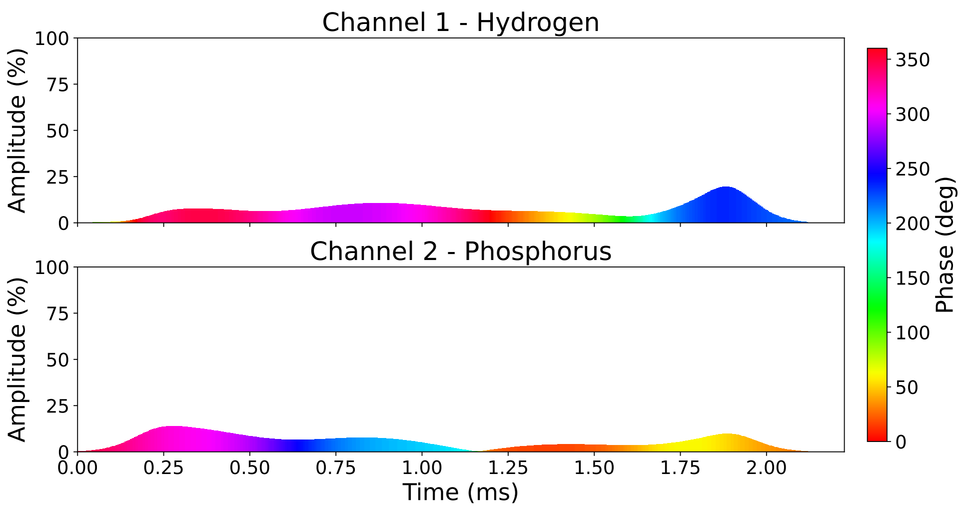

Figure 5.5: Pulse profile for the SWAP test optimization with a duration of 2.12 ms. Experimental fidelity of 70.1728%.

An experimental fidelity of around 70% was deemed insufficient for the purpose of this work, as it would not provide a confident measurement of the overlap between the input states.

5.2.2 Composite pulse optimization

The limited experimental performance of the single-pulse optimization suggested that the optimizer was converging to suboptimal solutions. To provide a more suitable initial guess, the destructive SWAP test was broken down into its component operations and optimized separately.

A CNOT gate was first optimized independently using the same optimization settings as in the previous study for durations ranging from 0.8 ms to 2 ms. The lower bound of 0.8 ms corresponds to $1/2J$ which is the time required for the J-coupling evolution described previously in Figure 4.2 to take effect. The resulting theoretical fidelities obtained for different pulse durations are shown in Figure 5.6.

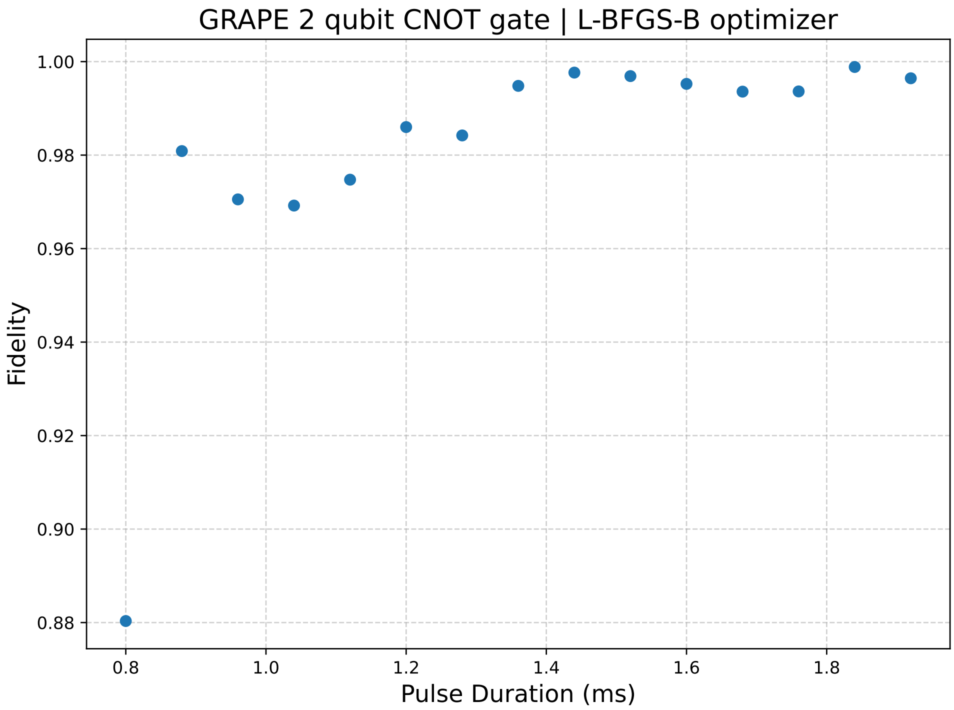

Figure 5.6: GRAPE-implemented CNOT gate theoretical fidelity for different pulse durations. Each data point represents the best theoretical fidelity obtained for a given pulse duration.

The selected pulse of duration 1.04 ms achieved an experimental fidelity of 87.13%, and its pulse profile is shown in Figure 5.7. Although the CNOT gate represents only a portion of the complete desired operation, this considerable increase in experimental fidelity is promising for the subsequent optimization of the complete destructive SWAP test operation. As it is a more complex optimization problem, this pulse sequence should provide a more accurate initial guess for the optimizer to refine.

As discussed in Section 4.1.1, single-qubit rotations can be implemented accurately using hard pulses. Accordingly, the Hadamard gate, which is equivalent to a sequence of single-qubit

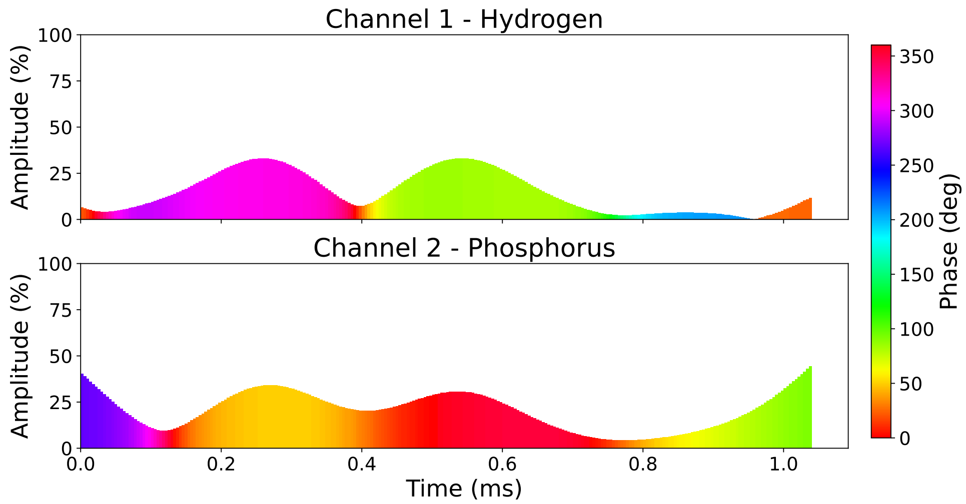

Figure 5.7: Pulse profile for the CNOT test optimization with a duration of 1.04 ms. Experimental fidelity of 87.13%. This pulse sequence was used as a portion of the initial guess for the composite pulse optimization of the complete destructive SWAP test operation.

rotations, was implemented as a hard-pulse $\pi/2$ rotation about the $\hat{x}$ axis followed by a π rotation about the $\hat{z}$ axis, and appended to the optimized CNOT pulse to form the initial guess for the composite pulse optimization of the complete destructive SWAP test operation.

The optimized composite pulse achieved a theoretical fidelity of 97.46% before convergence and an experimental fidelity of 89.66%, representing a substantial improvement over the direct optimization approach.

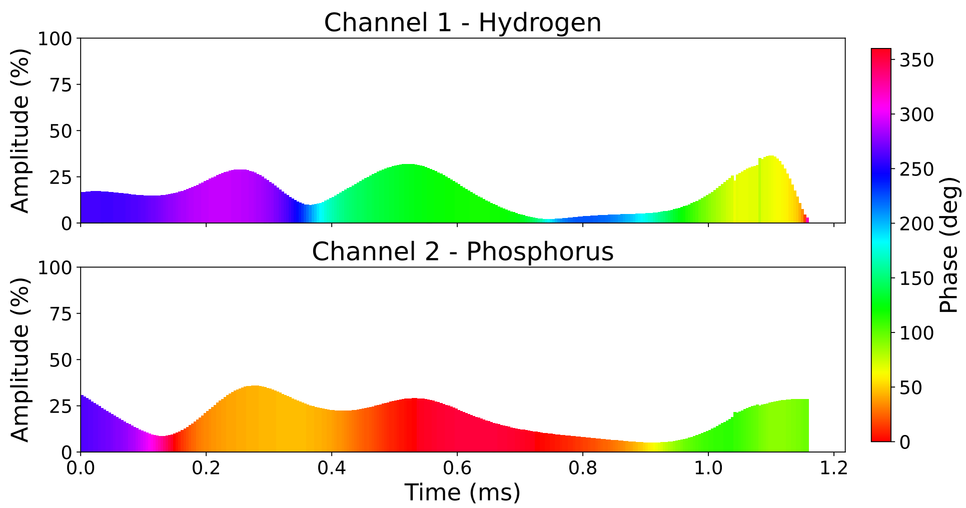

Figure 5.8: Pulse profile for the destructive SWAP test optimization with a composite initial guess. Experimental fidelity of 89.66%.

5.2.3 Gate characterization

Finally, to further characterize the performance of the destructive SWAP test control pulse, it was applied to multiple pairs of input states selected from the set $\{|0\rangle, |1\rangle, |+\rangle, |-\rangle\}$. For each input pair, the probability of measuring the outcome $|11\rangle$ was extracted, from which the overlap between the input states was calculated using:

$$|\langle\psi|\phi\rangle|^2 = 1 - 2P(|11\rangle) \tag{5.8}$$

The experimentally determined overlaps were then compared with their corresponding theoretical values. The experimental results reveal multiple distinct behaviors depending on the choice of input states. For example, when comparing identical states such as $|+\rangle$ and $|+\rangle$, or $|-\rangle$ and $|-\rangle$, the calculated overlaps are very close to the expected value of 1. In these cases, the small observed probability of measuring the state $|11\rangle$ can be attributed to thermal noise and other experimental imperfections, such as decoherence.

For orthogonal states such as $|+\rangle$ and $|-\rangle$, the measured probability of the $|11\rangle$ outcome slightly exceeded the expected value of 0.5, regardless of which state was assigned to the control qubit, as shown in Figures 5.9 and 5.10. On the other hand, for the state pair $|0\rangle$ and $|1\rangle$, the measured probability of $|11\rangle$ was consistently lower than expected. This discrepancy was more apparent when the control qubit was prepared in the state $|1\rangle$, for which the measured probability was approximately half the expected value. This is illustrated in Figures 5.11 and 5.12.

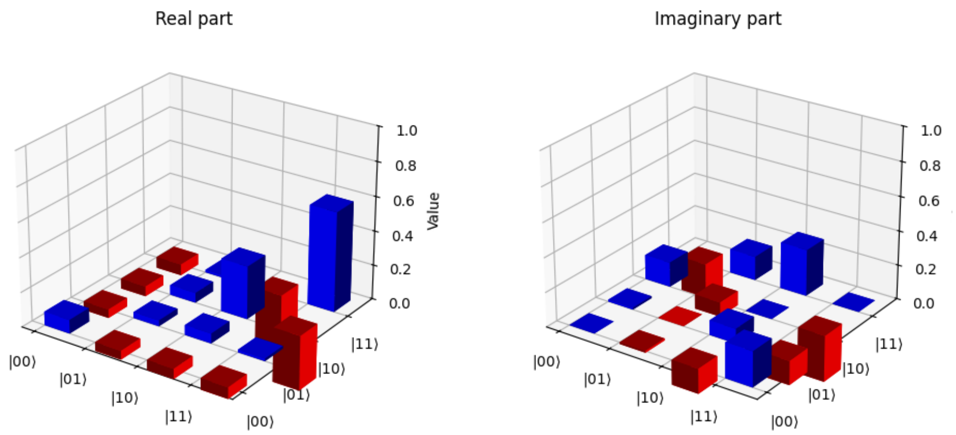

Figure 5.9: Experimental density matrix for the input state pair $|+\rangle$ and $|-\rangle$.

More significantly, for some input state pairs, exchanging the roles of the control and target qubits produced substantially different results. This behavior was observed for the state pairs $|1\rangle$ and $|+\rangle$, and $|0\rangle$ and $|-\rangle$, as illustrated by the experimental density matrices in Figures 5.15

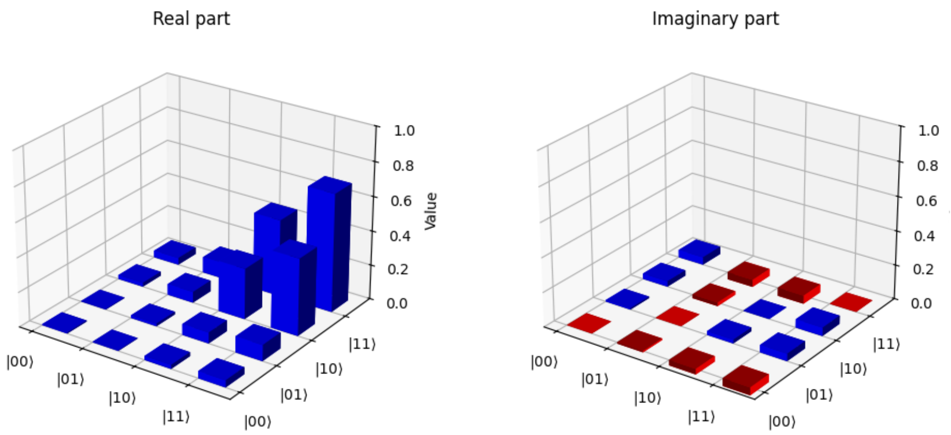

Figure 5.10: Experimental density matrix for the input state pair $|-\rangle$ and $|+\rangle$.

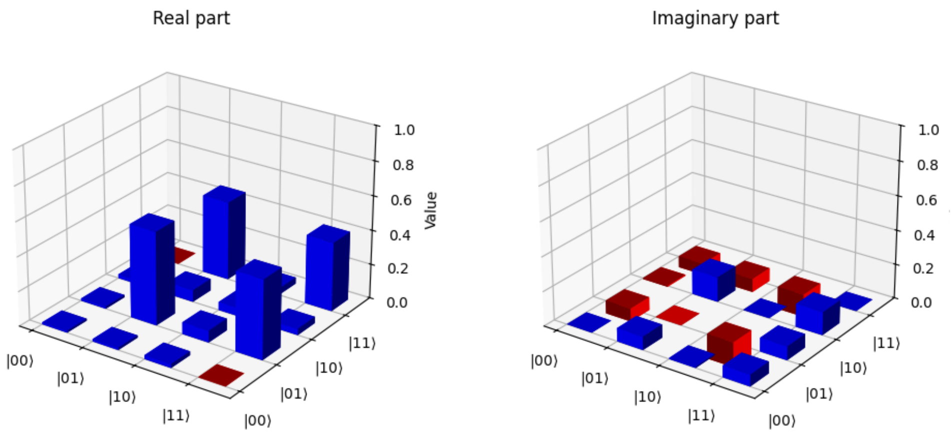

Figure 5.11: Experimental density matrix for the input state pair $|0\rangle$ and $|1\rangle$.

and 5.13. Since the ideal destructive SWAP test is symmetric with respect to the two input states, these results indicate that the implemented CNOT operation deviates from the ideal behavior. Consequently, further optimization of the CNOT pulse used to construct the initial guess for the composite pulse optimization is required.

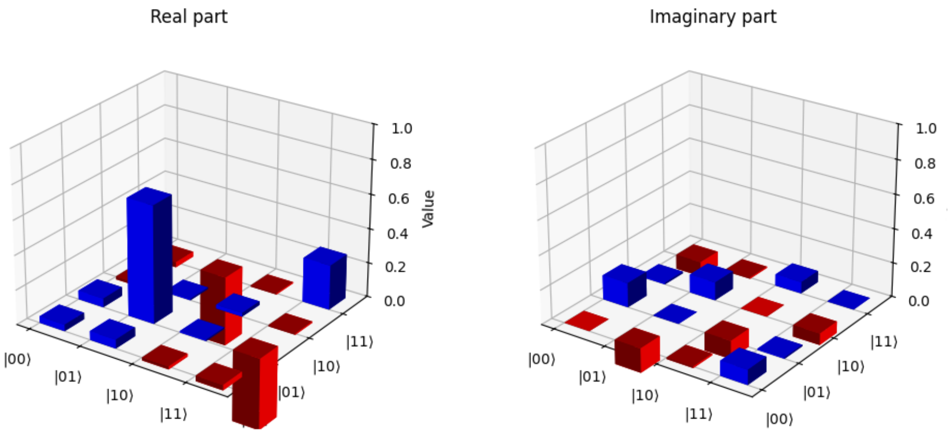

Figure 5.12: Experimental density matrix for the input state pair $|1\rangle$ and $|0\rangle$.

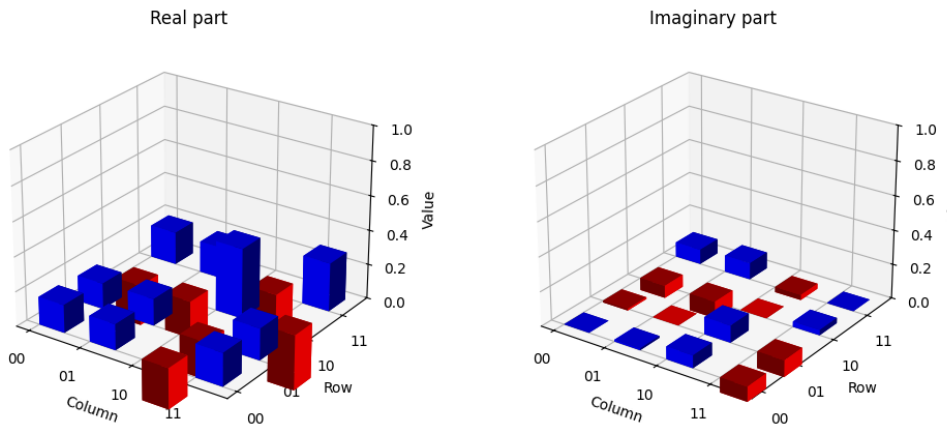

Figure 5.13: Experimental density matrix for the input state pair $|+\rangle$ and $|1\rangle$.

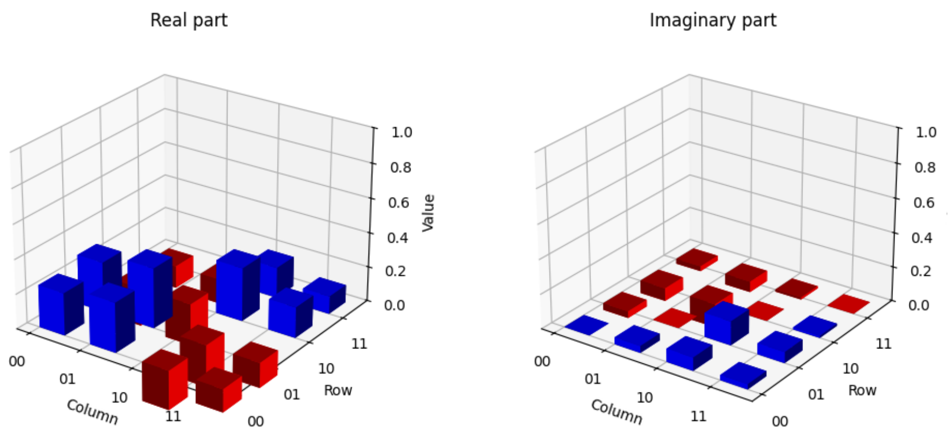

Figure 5.14: Experimental density matrix for the input state pair $|1\rangle$ and $|+\rangle$.

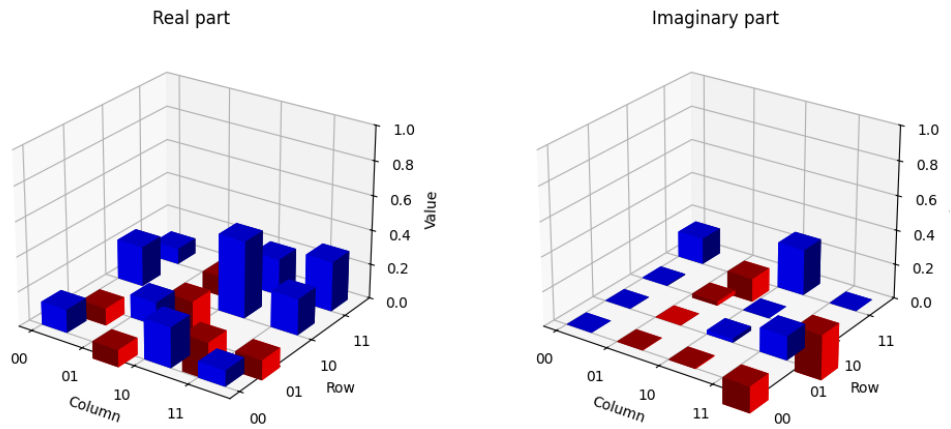

Figure 5.15: Experimental density matrix for the input state pair $|-\rangle$ and $|0\rangle$.

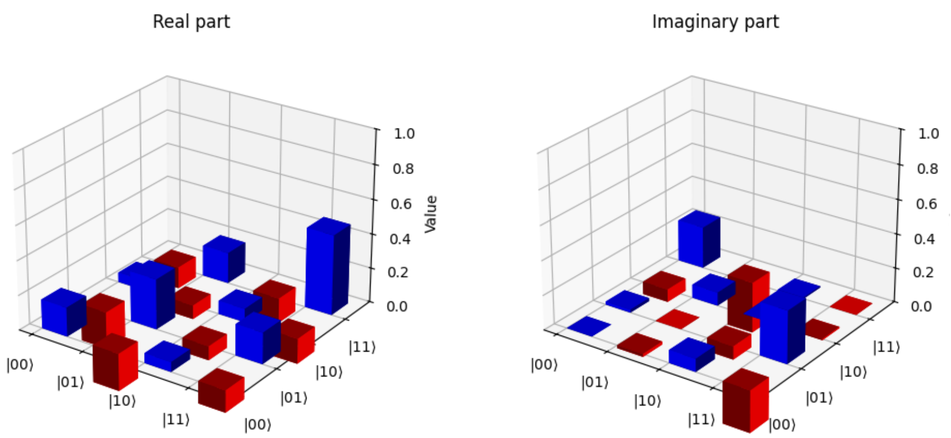

Figure 5.16: Experimental density matrix for the input state pair $|0\rangle$ and $|-\rangle$.

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Appendix A

Sample Appendix

Fusce mauris. Vestibulum luctus nibh at lectus. Sed bibendum, nulla a faucibus semper, leo velit ultricies tellus, ac venenatis arcu wisi vel nisl. Vestibulum diam. Aliquam pellentesque, augue quis sagittis posuere, turpis lacus congue quam, in hendrerit risus eros eget felis. Maecenas eget erat in sapien mattis porttitor. Vestibulum porttitor. Nulla facilisi. Sed a turpis eu lacus commodo facilisis. Morbi fringilla, wisi in dignissim interdum, justo lectus sagittis dui, et vehicula libero dui cursus dui. Mauris tempor ligula sed lacus. Duis cursus enim ut augue. Cras ac magna. Cras nulla. Nulla egestas. Curabitur a leo. Quisque egestas wisi eget nunc. Nam feugiat lacus vel est. Curabitur consectetuer.